

A Project Report

On

**“EARLY DETECTION OF LIFESTYLE DISORDERS THROUGH NERVE PULSES,
SLEEP AND STRESS INSIGHTS FROM WEARABLE WATCHES USING LIME AND
SHAP”**

Submitted in partial fulfillment to Jawaharlal Nehru Technological

University, of the requirements for the award of degree of

**Bachelor of Technology
in
Computer Science and Engineering**

by

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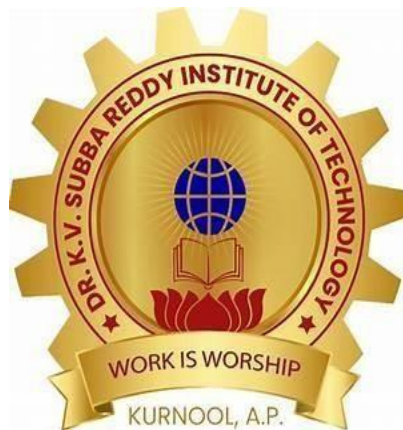
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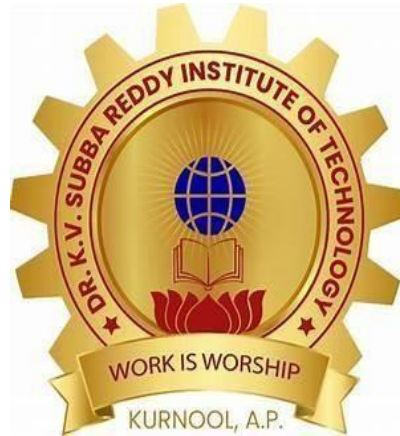
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CERTIFICATE

This is to certify that the Project report entitled **"EARLY DETECTION OF LIFESTYLE DISORDERS THROUGH NERVE PULSES, SLEEP AND STRESS INSIGHTS FROM WEARABLE WATCHES USING LIME AND SHAP"** is a bonafide record of work done by **P. KARTHIK (22FH1A0550), B. WASEEM FEROZ KHAN (22FH1A0528), G.KAMALAKAR (22FH1A0535), B SAI CHARAN TEJA(23FH5A0503)** at **Dr. K.V Subba Reddy Institute of Technology**, Kurnool, Affiliated to JNTUA, Anantapur during the year 2025-2026 in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**. This results presented in this has not been verified and found to be satisfactory. The result embodied in this Report has been submitted to any other University for the award of any other degree or diploma

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Abstract

Lifestyle-related health disorders such as hypertension, diabetes, cardiovascular stress, and sleep disorders are increasingly prevalent due to sedentary habits, stress, and irregular sleep cycles. Early detection of these conditions is essential for preventive healthcare. Wearable devices such as smartwatches continuously collect physiological data including pulse rate, heart rate variability, stress levels, and sleep patterns. This project proposes an intelligent machine learning–based system that analyzes nerve-system pulse features and sleep behavior data from wearable devices to predict potential lifestyle-related health risks at an early stage. The system integrates explainable AI techniques to improve transparency and trust in predictions. Experimental analysis demonstrates that combining pulse and sleep insights significantly enhances early risk detection and supports personalized health monitoring.

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Introduction

The project titled “**Predicting Lifestyle-Related Health Risks Using Pulse and Sleep Patterns**” presents an intelligent and data-driven healthcare system aimed at identifying potential health risks based on daily lifestyle indicators. With increasing lifestyle imbalances, parameters such as irregular sleep patterns and abnormal pulse rates have emerged as early signals of health deterioration. However, these indicators are often overlooked due to the lack of continuous monitoring and meaningful interpretation. This project addresses this gap by building a system that not only tracks such data but also converts it into actionable health insights.

The system collects user-specific inputs including pulse rate variations, sleep duration, sleep consistency, and related lifestyle attributes. These inputs are further enhanced through feature engineering techniques to extract meaningful patterns that improve prediction accuracy. By leveraging advanced machine learning algorithms such as XGBoost, Random Forest, and Decision Tree, the system analyzes complex relationships within the data to predict possible lifestyle-related health risks with high reliability.

A key strength of this project lies in the integration of Explainable Artificial Intelligence (XAI) techniques, specifically LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive exPlanations). Unlike traditional black-box models, this system provides clear explanations for each prediction, helping users and doctors understand the reasoning behind the results. This transparency builds trust and makes the system more suitable for real-world healthcare applications.

The platform is designed to serve multiple stakeholders. Patients can monitor their daily habits and receive early warnings about potential risks. Doctors are provided with structured reports and interpretable predictions, enabling better diagnosis and decision-making. Additionally, the system framework supports analytical insights that can be useful for healthcare management in identifying trends and improving overall service effectiveness.

This project goes beyond conventional healthcare applications by combining predictive analytics, feature engineering, and explainable AI into a unified solution. The use of intelligently generated features and optimized models ensures both accuracy and interpretability, making the system practical and impactful.

Overall, the project demonstrates how modern AI techniques can be applied to everyday health data to enable early detection, improve awareness, and promote preventive healthcare, ultimately contributing to a smarter and more proactive healthcare ecosystem.

Problem Statement

Lifestyle disorders often develop gradually and remain undetected until they reach severe stages. Traditional diagnostic methods rely on periodic clinical tests, which may fail to capture early warning signs. Although wearable devices generate continuous health data, this information is underutilized for proactive disease prediction. There is a need for an intelligent, automated system that can analyze wearable pulse and sleep data in real time and predict potential lifestyle-related health risks early, enabling timely intervention and preventive care.

Existing System

Current healthcare monitoring systems primarily focus on basic data collection and record maintenance rather than intelligent analysis. Popular health applications and wearable-integrated platforms collect parameters such as heart rate, step count, and sleep duration. However, these systems are mostly limited to displaying raw data in the form of graphs or summaries without deriving deeper insights.

Most existing solutions rely on static threshold-based alerts, where users are notified only when values exceed predefined limits. This approach lacks personalization and fails to consider individual variations in lifestyle patterns. Additionally, many machine learning-based healthcare systems function as black-box models, providing predictions without any explanation, making it difficult for users and doctors to trust or interpret the results.

Another limitation is the lack of integration between multiple health indicators. Existing systems often treat parameters like sleep and pulse independently, without analyzing their combined impact on overall health. Furthermore, trend analysis over time is either minimal or absent, preventing users from understanding how their habits are evolving and affecting their well-being.

Disadvantages of Existing System

- Lack of predictive capability; systems mainly focus on data display rather than early risk prediction
- Absence of explainability; predictions (if any) are not interpretable due to black-box models
- No use of Explainable AI techniques such as LIME and SHAP for understanding model decisions
- Poor integration of multiple health parameters like pulse rate and sleep patterns
- Limited or no trend analysis to track long-term lifestyle changes and health progression
- Generic threshold-based alerts instead of personalized health insights
- Underutilization of wearable device data for continuous and real-time analysis
- Lack of advanced feature engineering, leading to lower prediction accuracy
- No clear identification of key factors contributing to health risks
- Minimal support for doctors in terms of structured and interpretable patient reports
- Inability to convert raw health data into actionable insights for preventive care
- Fragmented system design with no unified platform connecting patients, doctors, and data analysis

Proposed System

The proposed system, **“Predicting Lifestyle-Related Health Risks Using Pulse and Sleep Patterns,”** is designed as an intelligent, predictive, and explainable healthcare platform that transforms daily lifestyle data into meaningful health insights. Unlike conventional systems, this platform focuses on early risk detection by analyzing the combined impact of physiological and behavioral parameters.

The system collects user data such as pulse rate trends, sleep duration, sleep consistency, and derived lifestyle indicators. These inputs are enhanced through advanced feature engineering to capture hidden patterns and relationships within the data. Machine learning algorithms including XGBoost, Random Forest, and Decision Tree are utilized to build robust predictive models capable of identifying potential health risks with high accuracy.

A major innovation in the proposed system is the integration of Explainable Artificial Intelligence techniques, specifically LIME and SHAP. These methods provide clear, feature-level

explanations for each prediction, allowing users and doctors to understand why a particular risk is identified. This transparency ensures trust, reliability, and practical usability in real-world healthcare scenarios.

The system is also designed to support continuous monitoring and trend analysis. By integrating data from wearable devices such as smartwatches, it enables real-time data collection and long-term tracking of user behavior. This helps in identifying gradual lifestyle changes and their impact on health.

Additionally, the platform provides a structured interface for different stakeholders. Patients receive personalized insights and early warnings, doctors can access interpretable reports for better diagnosis, and the system framework supports analytical insights that can be extended for healthcare management and decision-making.

Advantages of Proposed System

- Enables early prediction of lifestyle-related health risks using advanced machine learning models
- Provides explainable predictions using LIME and SHAP, increasing transparency and trust
- Combines multiple parameters such as pulse and sleep for more accurate and meaningful analysis
- Utilizes advanced feature engineering to improve model performance and prediction accuracy
- Supports continuous monitoring through integration with wearable devices
- Performs effective trend analysis to track long-term health patterns and lifestyle changes
- Delivers personalized insights instead of generic threshold-based alerts
- Identifies key contributing factors behind each prediction for better understanding
- Assists doctors with structured, interpretable patient reports for improved decision-making
- Converts raw health data into actionable insights for preventive healthcare
- Scalable and adaptable system suitable for real-world healthcare applications
- Bridges the gap between data collection and intelligent health analysis.

Objectives

- To develop a system that predicts lifestyle-related health risks using pulse rate and sleep pattern data
- To apply machine learning algorithms such as XGBoost, Random Forest, and Decision Tree for accurate health risk prediction
- To implement Explainable AI techniques (LIME and SHAP) to provide clear and interpretable prediction results
- To perform effective feature engineering to extract meaningful patterns from raw health data
- To enable continuous health monitoring by integrating data from wearable devices like smartwatches
- To analyze trends in user lifestyle over time and provide personalized insights for preventive healthcare

Chapter 1: Literature Survey

1. Title: Wearable Sensor-Based Health Monitoring Systems

Author: J. A. Patel, R. Kumar

Abstract:

This study reviews wearable sensor technologies for continuous health monitoring and highlights their role in early detection of lifestyle disorders.

2. Title: Machine Learning for Lifestyle Disease Prediction

Author: S. Banerjee, M. Das

Abstract:

The authors explore machine learning models for predicting lifestyle diseases using physiological and behavioral data, showing improved prediction accuracy.

3. Title: Sleep Pattern Analysis for Health Risk Assessment

Author: L. Zhang, H. Liu

Abstract:

This research demonstrates the importance of sleep quality metrics in predicting long-term health risks such as cardiovascular and metabolic disorders.

4. Title: Pulse Signal Analysis for Stress and Health Monitoring

Author: A. Sharma, N. Gupta

Abstract:

The paper focuses on analyzing pulse and heart rate variability to assess stress levels and nervous system health.

5. Title: Explainable AI in Healthcare Applications

Author: M. Ribeiro, S. Singh

Abstract:

This study emphasizes the role of explainable AI techniques in healthcare systems, improving transparency, trust, and clinical adoption of machine learning models.

Chapter 2: System Requirements

2.1 Hardware Requirements

- System : Pentium Iv 2.4 Ghz.
- Hard Disk : 40 Gb.
- Ram : 512 Mb.

2.2 Software Requirements

Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python
- Framework: Django (for backend development)
- Frontend Technologies: HTML, CSS, JavaScript
- Database: SQLite / PostgreSQL (Neon DB for cloud usage)
- Libraries & Tools:
 - Machine Learning: Scikit-learn, XGBoost
 - Data Processing: Pandas, NumPy
 - Visualization: Matplotlib, Seaborn
 - Explainable AI: LIME, SHAP
- API Integration: For wearable data
- Version Control: Git & GitHub

2.1 System Architecture

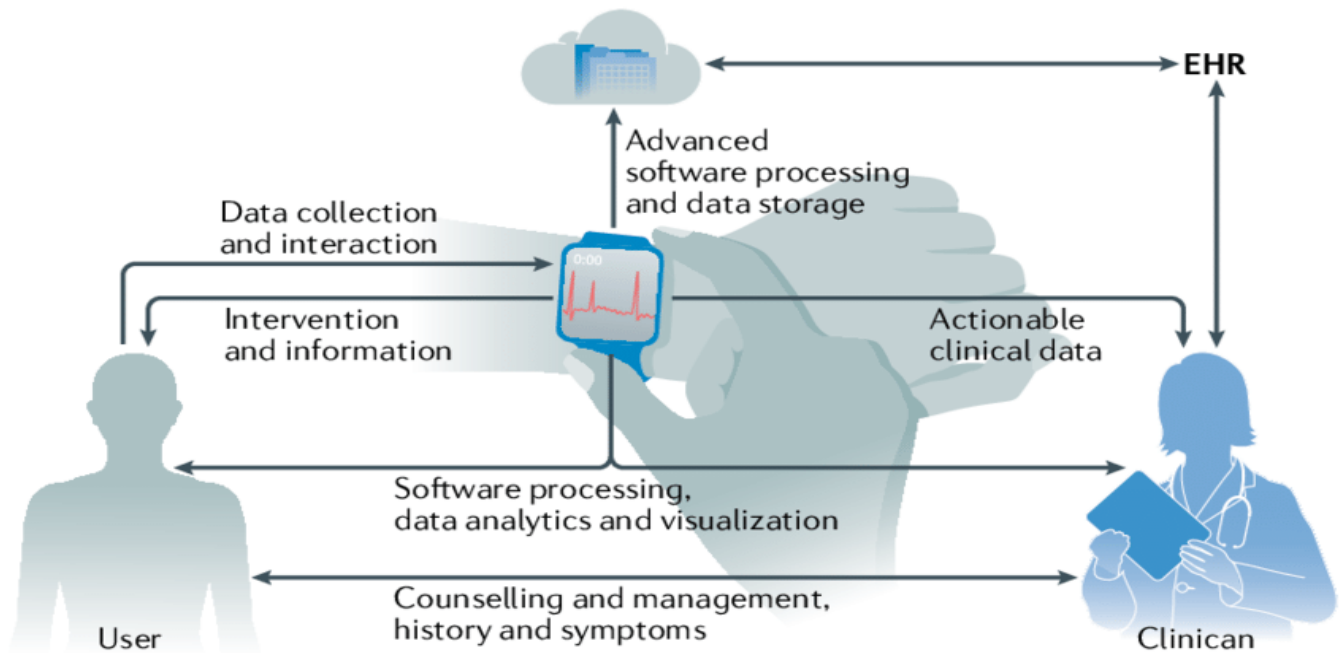


Fig: 2.1 System Architecture

The system architecture is designed as a multi-layered structure that ensures smooth data flow from collection to prediction and visualization. At the initial stage, user health data such as pulse rate and sleep patterns are collected either manually or through wearable devices like smartwatches. This data is transmitted to the processing layer, where it is cleaned, structured, and enhanced using feature engineering techniques.

The processed data is stored in a centralized database, enabling efficient access and management. In the next layer, machine learning models such as XGBoost, Random Forest, and Decision Tree analyze the data to predict potential lifestyle-related health risks. To ensure transparency, Explainable AI techniques like LIME and SHAP are applied, providing clear insights into how predictions are made.

The application layer provides an interactive interface for users and doctors, where patients can track their health trends and doctors can view detailed reports and predictions. Visualization tools are used to present data in an understandable format. The system also supports integration with external platforms like wearable devices and can be extended to electronic health records. Overall, the architecture enables continuous monitoring, intelligent prediction, and effective communication between all components.

Chapter 3: Feasibility Study

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- Economical Feasibility
- Technical Feasibility
- Social Feasibility

3.1 Economical Feasibility

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

3.2 Technical Feasibility

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

3.3 Social Feasibility

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it.

Chapter 4: Modules Description

1. Data Acquisition Module

This module is responsible for collecting physiological and lifestyle data from users. It gathers pulse rate, sleep duration, sleep quality, activity levels, and basic demographic information through wearable devices, mobile sensors, or user inputs. The collected raw data forms the foundation for health risk prediction.

2. Data Preprocessing Module

The preprocessing module cleans and prepares the collected data for analysis. It handles missing values, removes noise from pulse signals, normalizes sleep metrics, and converts raw sensor data into structured formats. This step ensures data consistency, reliability, and improved model accuracy.

3. Feature Extraction Module

This module extracts meaningful features from pulse and sleep data. Pulse-related features include average heart rate, resting heart rate, and heart rate variability, while sleep-related features include total sleep time, sleep efficiency, and sleep pattern irregularities. These features represent lifestyle behaviors linked to health risks.

4. Lifestyle Pattern Analysis Module

The lifestyle pattern analysis module identifies trends and behavioral patterns in pulse and sleep data. It analyzes daily routines, detects irregular sleep schedules, prolonged elevated pulse rates, and lifestyle inconsistencies that may indicate unhealthy habits or stress-related conditions.

5. Health Risk Prediction Module

This core module uses machine learning or deep learning algorithms to predict potential lifestyle-related health risks such as stress disorders, cardiovascular risks, sleep disorders, or fatigue-related issues. Based on extracted features, the model classifies users into different risk categories (low, medium, high).

6. Recommendation and Alert Module

Based on predicted risks, this module generates personalized health recommendations and alerts. It suggests lifestyle improvements such as better sleep schedules, stress management practices, physical activity adjustments, or medical consultation alerts for high-risk users.

7. Visualization and Reporting Module

This module presents analyzed data and prediction results in an easy-to-understand visual format. Dashboards, charts, and trend graphs display pulse variations, sleep patterns, and risk levels, helping users and clinicians quickly interpret health insights.

8. User and Clinician Interface Module

This module provides separate interfaces for users and healthcare professionals. Users can view their health status and recommendations, while clinicians can access summarized reports, risk assessments, and historical data to support monitoring and decision-making.

9. Data Storage and Security Module

This module securely stores user data, extracted features, and prediction results. It ensures privacy through encryption, access control, and compliance with healthcare data protection standards, supporting long-term health monitoring and analysis.

Chapter 5: Algorithms Used

The proposed system utilizes multiple machine learning algorithms to ensure accurate and reliable prediction of lifestyle-related health risks. Each algorithm is selected based on its ability to handle structured health data and capture complex patterns effectively. **Moving Average Filter** – Smooths short-term fluctuations in pulse rate data.

XGBoost (Extreme Gradient Boosting)

XGBoost is one of the main algorithms used in this project for predicting lifestyle-related health risks from pulse and sleep pattern data. It is an advanced ensemble learning technique that builds a sequence of decision trees, where each new tree learns from the errors made by the previous trees. In this project, XGBoost is highly suitable because the relationship between pulse, sleep duration, sleep irregularity, and generated health-related features is not always simple or linear. The algorithm is capable of identifying these hidden interactions effectively. It also performs well when the dataset contains a mixture of direct and derived features generated during preprocessing and feature engineering. Another important reason for selecting XGBoost is its strong prediction accuracy and ability to control overfitting through regularization. Since healthcare-related predictions require reliability, this algorithm helps in producing more stable and dependable results. XGBoost also provides feature importance scores, which are useful in identifying which health factors influence the prediction most. These strengths make it highly effective for detecting potential risks at an early stage. In this project, it plays a major role in improving the overall intelligence and accuracy of the proposed system. **Heart Rate Variability (HRV) Analysis** Extracts time-domain and frequency-domain features such as RMSSD, SDNN, and LF/HF ratio.

Extracts time-domain and frequency-domain features such as RMSSD, SDNN, and LF/HF ratio.

• Random Forest

Random Forest is another important algorithm used in this project to predict health risks based on user lifestyle data. It is an ensemble technique that creates multiple decision trees using different subsets of training data and combines their outputs to produce the final prediction. In the context of this project, Random Forest is useful because health data collected from pulse and sleep patterns may contain variations and inconsistencies between different users. By combining the decisions of many trees, the algorithm reduces the possibility of inaccurate predictions caused by noisy or irregular data. It is capable of handling multiple input features at the same time and can identify important patterns that may not be visible through a single decision path. Random Forest is

also less prone to overfitting compared to a normal decision tree, which makes it more reliable for practical healthcare analysis. It supports better generalization when the system is tested on unseen data. In this project, the algorithm helps in generating balanced and robust predictions by considering a wide range of possible feature combinations. It also assists in determining feature importance, which supports analysis of how pulse and sleep-related parameters influence health risk. Therefore, Random Forest strengthens the system by improving consistency, robustness, and predictive performance.

• **Decision Tree**

Decision Tree is used in this project as a simple and interpretable machine learning algorithm for predicting lifestyle-related health risks. It works by splitting the data into branches based on conditions applied to different input features such as pulse values, sleep duration, and other generated indicators. Each split represents a decision rule, and the final output is obtained at the leaf node. In this project, Decision Tree is useful because it clearly shows how the system arrives at a particular prediction. This makes it easier to understand the influence of specific health parameters on the final result. For example, the model can reveal how abnormal pulse trends combined with poor sleep patterns may lead to a higher predicted risk. Its transparent structure makes it especially helpful during analysis, testing, and comparison with more advanced models. Although it may not always provide the same level of accuracy as ensemble methods, it offers strong interpretability, which is valuable in healthcare applications. It can also serve as a baseline model to evaluate the improvement achieved by XGBoost and Random Forest. In this project, Decision Tree contributes to the explainability and simplicity of the system while still supporting effective prediction. Its rule-based nature aligns well with the need to make healthcare predictions understandable to both developers and end users.

• **LIME (Local Interpretable Model-Agnostic Explanations)**

LIME is used in this project as an Explainable AI technique to provide local explanations for individual predictions generated by the machine learning models. In healthcare applications, prediction alone is not sufficient because users and doctors also need to know why a particular result was produced. LIME addresses this requirement by explaining the contribution of input features for a single prediction instance. In this project, after the system predicts a lifestyle-related health risk, LIME highlights the specific factors that influenced that result, such as poor sleep duration, unstable pulse values, or important engineered features derived from user data. This helps in making the output more transparent and understandable. One of the major advantages of LIME is

that it can work with different machine learning models without depending on their internal structure. This makes it suitable for explaining predictions from XGBoost, Random Forest, and Decision Tree within the same system. It improves user trust because the patient or doctor can see the reasons behind the risk level instead of only viewing a final label. In this project, LIME plays an important role in transforming the model from a black-box predictor into a more interpretable healthcare assistant. As a result, it supports better decision-making and increases confidence in the system's predictions.

- **SHAP (SHapley Additive exPlanations)**

SHAP is another Explainable AI technique integrated into this project to provide both local and global interpretation of machine learning predictions. It is based on the concept of Shapley values, where each feature is assigned a contribution value representing its impact on the final output. In this project, SHAP is used to understand how pulse-related, sleep-related, and engineered features affect lifestyle-related health risk prediction. For an individual user, SHAP can show whether a specific feature increased or decreased the predicted risk. At the overall system level, it can identify which features consistently have the strongest influence across the entire dataset. This is highly valuable in healthcare because it provides a deeper understanding of model behavior and supports transparency. SHAP is especially effective when used with XGBoost because it can explain complex patterns while maintaining clarity in the output. In this project, it helps in identifying major health indicators that contribute most to the final decision, which can be useful for both medical interpretation and model validation. Compared to simple feature importance, SHAP gives more detailed and reliable insights. Therefore, SHAP strengthens the explainability, trustworthiness, and analytical value of the system, making the prediction process more meaningful and scientifically interpretable.

Performance Evaluation Metrics

To evaluate the effectiveness of the prediction models used in this project, several performance metrics are applied. These metrics help in measuring how accurately the system predicts lifestyle-related health risks based on pulse and sleep pattern data.

- **Confusion Matrix**

The confusion matrix is used to visualize the performance of the classification models. It provides a detailed breakdown of predictions by comparing actual values with predicted values. In this project, it helps in identifying how many instances are correctly predicted as risk or no-risk, and

how many are misclassified. This gives a clear understanding of model strengths and weaknesses, especially in detecting false positives and false negatives, which are critical in healthcare scenarios.

- **Accuracy**

Accuracy measures the overall correctness of the model by calculating the proportion of correctly predicted instances out of total predictions. In this project, it indicates how well the model classifies health risk conditions based on the given input data. Although accuracy provides a general idea of performance, it is not always sufficient when dealing with imbalanced healthcare data.

- **Precision**

Precision measures how many of the predicted positive cases are actually correct. In this system, it reflects how accurately the model identifies users who are at risk. High precision ensures that when the system predicts a health risk, it is reliable and not producing too many false alarms.

- **Recall (Sensitivity)**

Recall measures how many actual positive cases are correctly identified by the model. In this project, it is important because missing a real health risk (false negative) can be dangerous. A higher recall ensures that most of the risky cases are detected by the system.

- **F1-Score**

F1-Score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance. In this project, it is used to ensure that both false positives and false negatives are minimized. It gives a better evaluation compared to accuracy when the dataset is not perfectly balanced.

Chapter 6: UML Diagrams

UML (Unified Modeling Language) is used in this project as a standardized approach to visually represent the design and structure of the healthcare prediction system based on pulse and sleep patterns. It helps in clearly illustrating how different components of the system such as user data collection, machine learning models, Explainable AI modules, and interfaces interact with each other.

In this project, UML plays an important role in modeling the flow of data from wearable devices or user inputs to the prediction system and finally to the user and doctor interfaces. It provides a clear visualization of how health data is processed, how predictions are generated using algorithms like XGBoost, Random Forest, and Decision Tree, and how explanations are provided using LIME and SHAP.

The use of UML diagrams helps in simplifying complex system architecture by representing it in a graphical format. It supports better understanding of modules such as data preprocessing, feature engineering, prediction, and visualization dashboards. This makes it easier to communicate the system design to developers, evaluators, and stakeholders.

UML is independent of programming languages, which makes it suitable for representing this project regardless of the technologies used such as Python and Django. It also supports scalability, allowing future enhancements like integration with advanced wearable devices or healthcare systems.

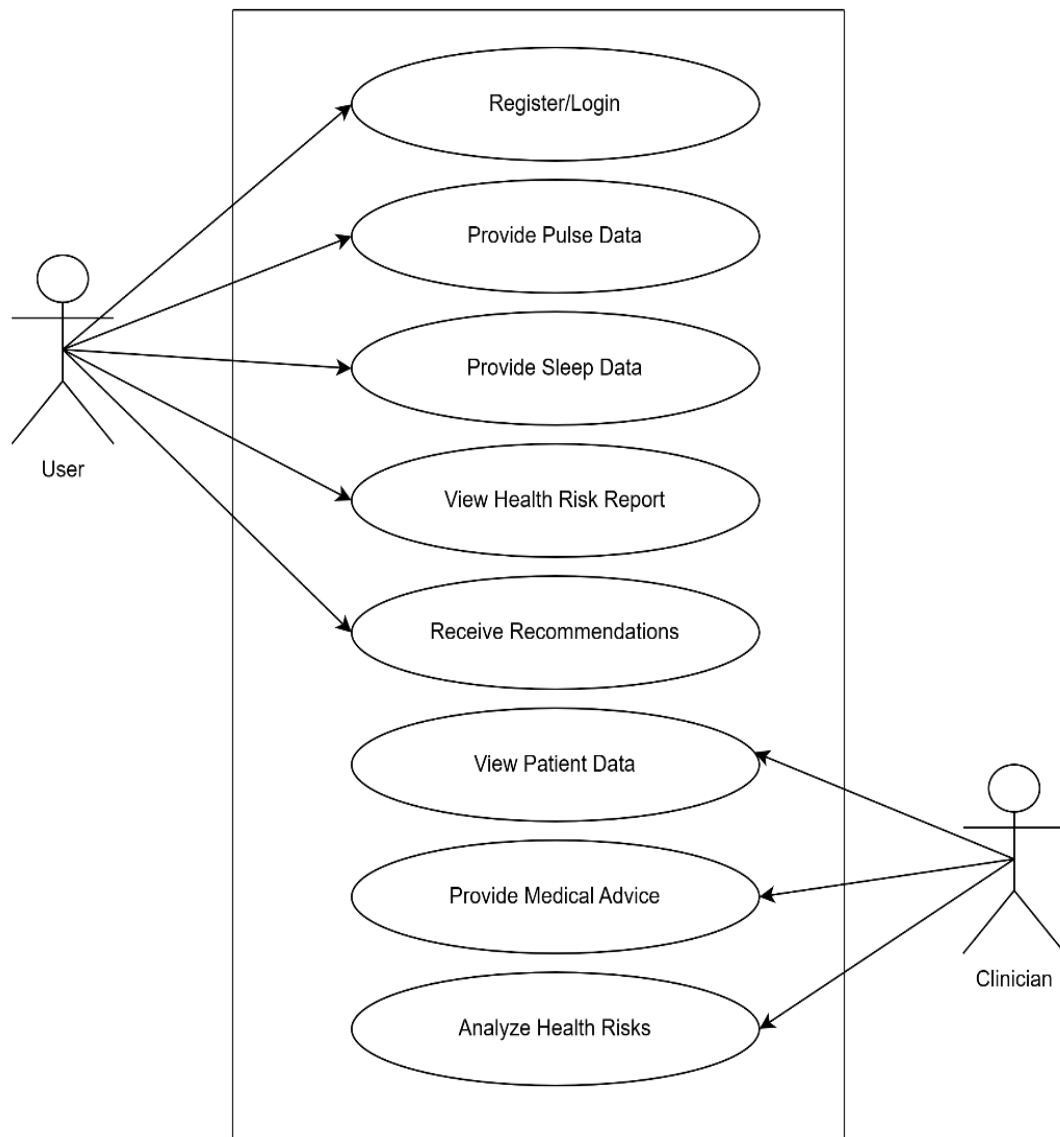
Goals of UML in This Project

1. To provide a clear visual representation of the healthcare system architecture and workflow
2. To model interactions between users, wearable devices, machine learning models, and dashboards
3. To represent system modules such as data processing, prediction, and Explainable AI clearly
4. To simplify understanding of complex processes like health risk prediction and trend analysis
5. To ensure structured and organized system design for development and documentation
6. To support future scalability and integration with advanced healthcare technologies
7. To improve communication between developers, evaluators, and stakeholders

Use Case Diagram

In this project, the Use Case Diagram represents the interaction between different actors such as **Patient, Doctor, and System/Admin** with the healthcare prediction system. It shows how patients input pulse and sleep data (manually or via wearable devices), view predictions, and track health trends. Doctors can access patient reports, analyze predictions, and understand explanations generated

using LIME and SHAP. The diagram clearly illustrates the functionalities provided by the system, such as data entry, prediction generation, visualization, and report access, making it easy to understand how each user interacts with the system..

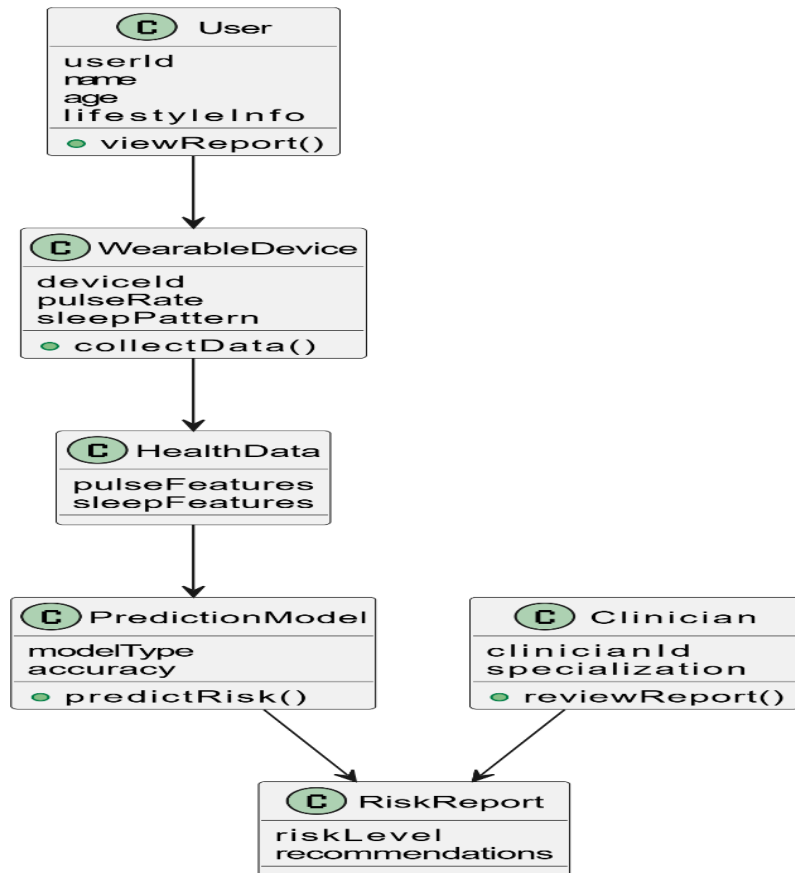


6.1 Use Case Diagram

Class Diagram

The Class Diagram describes the structural design of the system by showing different classes such as User, HealthData, PredictionModel, FeatureEngine, and Report. Each class contains attributes like pulse rate, sleep duration, and derived features, along with methods for processing and prediction. It also represents relationships between classes, such as how user data is linked to

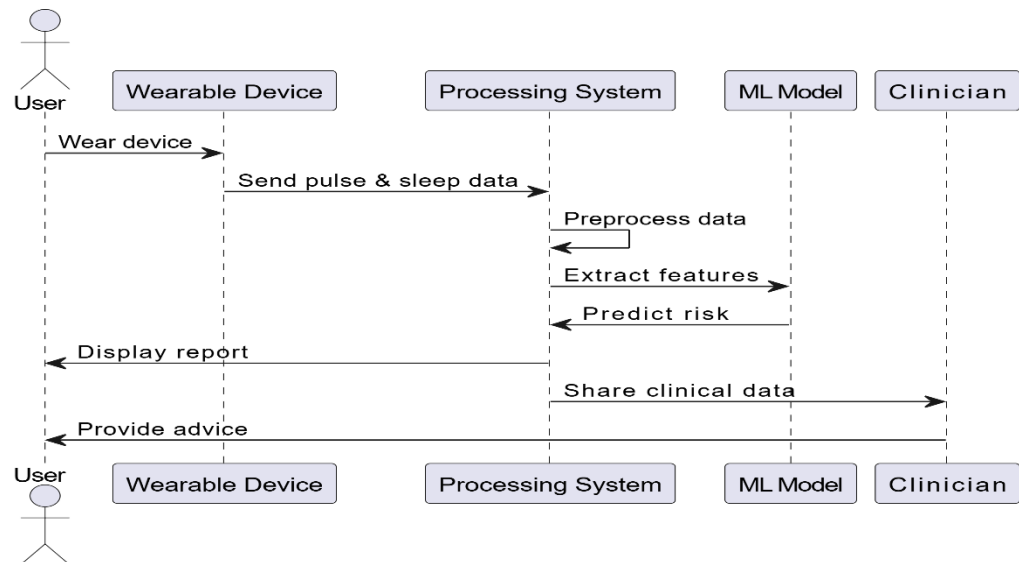
prediction models and how results are stored and retrieved. This diagram helps in understanding how the system is organized internally and how data flows between components.



6.2 Class Diagram

Sequence Diagram

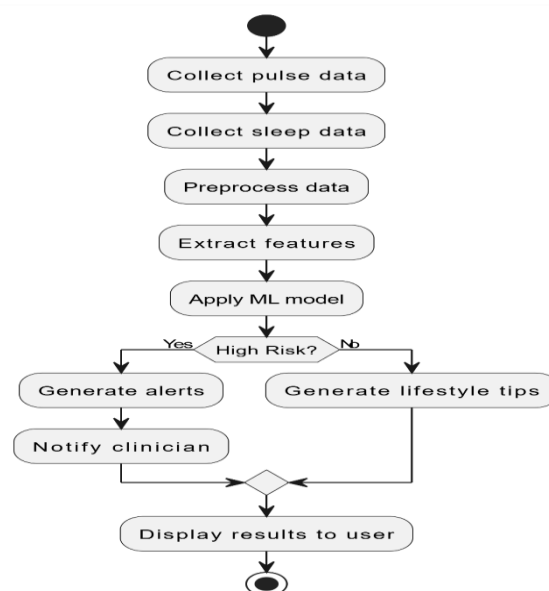
The Sequence Diagram illustrates the step-by-step interaction between system components over time. In this project, it shows how user data is collected, sent to the backend, processed through feature engineering, and then passed to machine learning models for prediction. After prediction, LIME and SHAP generate explanations, and the results are displayed to the user or doctor. This diagram highlights the exact order of operations and communication between modules, ensuring clarity in system workflow.



6.3 Sequence Diagram

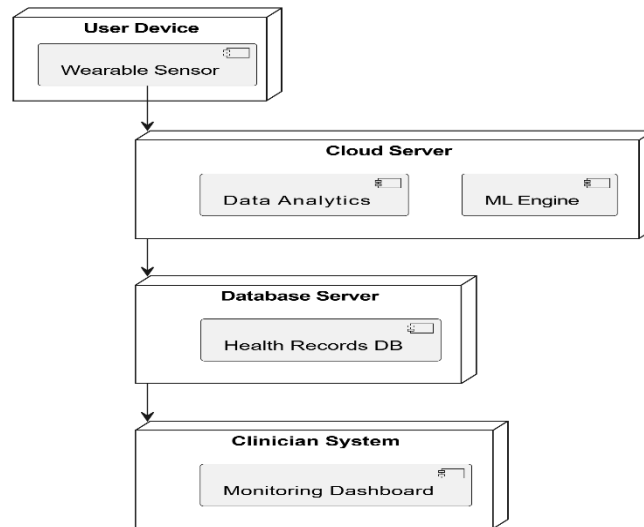
Activity Diagram

The Activity Diagram represents the complete workflow of the system from data input to final output. It includes steps such as data collection, preprocessing, feature generation, prediction, explanation, and result visualization. It also shows decision points, such as whether the predicted risk is high or low, and how the system responds accordingly. This diagram helps in understanding the overall functioning and logic flow of the healthcare prediction system.



6.4 Activity Diagram

Deployment Diagram



6.5 Deployment Diagram

The Deployment Diagram illustrates how the system is physically deployed. In this project, it shows components such as the user device (mobile or laptop), wearable devices (smartwatch), web server (Django application), database (Neon DB/SQLite), and cloud services. It explains how data flows between these components and how the system operates in a real-world environment. This diagram ensures clarity in system infrastructure and deployment setup.

Chapter 7: Functional Requirements

User Registration and Authentication

The system shall allow users to register and securely log in using valid credentials.

Pulse Data Collection

The system shall collect real-time or periodic pulse rate data from wearable devices or sensors.

Sleep Data Collection

The system shall record sleep duration, sleep quality, and sleep patterns from user inputs or wearable devices.

Data Preprocessing

The system shall clean, normalize, and filter raw pulse and sleep data to remove noise and inconsistencies.

Feature Extraction

The system shall extract meaningful health indicators such as average heart rate, heart rate variability, sleep efficiency, and sleep irregularities.

Lifestyle Pattern Analysis

The system shall analyze user lifestyle patterns based on pulse and sleep behavior over time.

Health Risk Prediction

The system shall predict lifestyle-related health risks using machine learning or deep learning models.

Risk Classification

The system shall classify users into risk categories such as low, medium, or high risk.

Recommendation Generation

The system shall provide personalized lifestyle recommendations based on predicted risk levels.

Alert and Notification

The system shall generate alerts for users and clinicians when high-risk conditions are detected.

Visualization and Reporting

The system shall display health reports, graphs, and trend analysis through an interactive dashboard.

Clinician Access

The system shall allow clinicians to view patient risk summaries and provide medical feedback.

Data Storage and Management

The system shall store user data, extracted features, and prediction results securely for future analysis.

7.2 Non-Functional Requirements

1. Performance Requirements

- The system shall process user data and generate predictions within an acceptable response time.
- The system shall support multiple users simultaneously without performance degradation.

2. Security Requirements

- The system shall ensure secure data transmission using encryption.
- The system shall restrict access to sensitive health data based on user roles.
- The system shall protect user privacy in compliance with healthcare data standards.

3. Reliability Requirements

- The system shall provide accurate and consistent predictions.
- The system shall recover gracefully from system failures without data loss.

4. Scalability Requirements

- The system shall handle increasing amounts of user data and wearable device inputs.
- The system shall support future expansion of additional health parameters.

5. Usability Requirements

- The system shall provide a user-friendly interface understandable by non-technical users.
- Health insights shall be presented in simple, visual formats.

6. Maintainability Requirements

- The system shall be modular to allow easy updates of algorithms or features.
- The system shall support debugging and performance monitoring.

7. Availability Requirements

- The system shall be available for use 24/7 except during scheduled maintenance.
- Downtime shall be minimized to ensure continuous health monitoring.

8. Compatibility Requirements

- The system shall be compatible with different wearable devices and operating platforms.
- The system shall support web and mobile-based access.

9. Accuracy Requirements

- The prediction models shall achieve high accuracy with minimal false positives and negatives.
- The system shall periodically retrain models to improve prediction quality.

Chapter 8: System Environment

TECHNOLOGIES USED IN THE PROJECT

The system environment defines the overall setup in which the healthcare prediction system is developed, deployed, and executed. This project is designed to operate in a flexible and scalable environment, supporting both local development and cloud-based deployment.

The development environment consists of Python and Django for backend processing, along with HTML, CSS, JavaScript, and Bootstrap for building an interactive user interface. Machine learning models such as XGBoost, Random Forest, and Decision Tree are integrated using libraries like Scikit-learn and XGBoost, while data processing is handled using Pandas and NumPy. Explainable AI techniques (LIME and SHAP) are incorporated to provide transparency in predictions.

The system runs on standard operating systems such as Windows or Linux and is accessible through web browsers like Chrome or Edge. For data storage, databases such as SQLite (for local use) and Neon DB (for cloud-based storage) are used. Cloudinary is utilized for handling media and external storage needs.

In terms of execution environment, the system supports integration with wearable devices such as smartwatches, enabling real-time collection of pulse and sleep data. The application is deployed as a web-based platform, allowing users and doctors to access it from any device with internet connectivity.

Overall, the system environment ensures smooth functioning, scalability, and accessibility while supporting advanced features like machine learning predictions, real-time monitoring, and explainable AI.

1. Python (Project-Specific Usage)

Python is used as the core implementation language for the proposed system. In this project, Python plays a central role in integrating wearable data processing, machine learning model execution, explainable artificial intelligence, and web-based application development into a single cohesive platform. The wearable data collected in the form of pulse rate, heart rate variability, sleep duration, sleep quality score, stress level, and activity indicators is processed using Python to convert raw inputs into structured and meaningful features suitable for prediction. Python is extensively used for data preprocessing operations such as cleaning missing values, scaling numerical features, encoding categorical attributes, and preparing datasets for training and testing. In addition, Python enables seamless training, storage, and deployment of machine learning models and supports

explainable AI techniques that help interpret prediction results. Its stability, flexibility, and rich ecosystem make it well suited for healthcare-oriented predictive systems that require accuracy, transparency, and reliability.

Python Libraries Used and Their Purpose

• NumPy

NumPy is used as the core numerical computing library in this project. It enables efficient handling of large-scale health data through optimized array operations. During preprocessing, NumPy is used to perform mathematical transformations on pulse rate values, sleep duration, and derived features such as averages, variations, and ratios. Since machine learning models require numerical input in structured form, NumPy ensures fast and memory-efficient data representation. It also supports vectorized operations, which significantly reduce computation time when working with multiple health parameters simultaneously. In this project, NumPy acts as the foundation for data manipulation before it is passed to higher-level libraries like Pandas and Scikit-learn.

• Pandas

Pandas plays a major role in organizing and preparing the dataset used for prediction. It is used to create structured DataFrames containing user inputs such as pulse readings, sleep hours, and engineered features. In this project, Pandas is responsible for handling missing values, filtering inconsistent records, and transforming raw wearable or manual data into a clean format. It is also used extensively for feature engineering, where new meaningful attributes are generated from existing data, such as sleep consistency, pulse variation, or combined health indicators. These engineered features directly improve the model's predictive performance. Pandas also simplifies data analysis by enabling easy slicing, grouping, and aggregation of health records.

• Scikit-learn

Scikit-learn is used as the primary library for implementing machine learning workflows in this project. It provides tools for splitting the dataset into training and testing sets, ensuring that the model is evaluated properly. Algorithms like Decision Tree and Random Forest are implemented using this library. It also supports preprocessing techniques such as scaling and encoding if required. One of its most important roles in this project is performance evaluation, where it calculates metrics like accuracy, precision, recall, and F1-score. These metrics help in validating how well the system

predicts lifestyle-related health risks. Scikit-learn ensures that the entire machine learning pipeline is structured, consistent, and reliable.

- **XGBoost**

XGBoost is used as the main prediction algorithm due to its high efficiency and superior performance in handling structured data. It works by building multiple trees sequentially and correcting errors from previous iterations, which makes it highly accurate. In this project, XGBoost is particularly effective in capturing complex relationships between pulse patterns, sleep behavior, and engineered features. It also includes regularization techniques that prevent overfitting, making the model more reliable for real-world health predictions. Another important advantage is its ability to provide feature importance, which helps in understanding which health factors contribute most to the prediction. XGBoost significantly enhances the intelligence and accuracy of the system.

- **Matplotlib**

Matplotlib is used for generating visual representations of health data and prediction results. In this project, it is used to plot graphs such as pulse trends over time, sleep pattern variations, and comparison charts for predicted risk levels. These visualizations help users and doctors easily interpret complex data. By converting numerical outputs into graphical formats, Matplotlib improves the usability of the system. It is also useful during development and testing phases to analyze model performance and data distribution.

- **Seaborn**

Seaborn is used as an advanced visualization library built on top of Matplotlib. It is mainly used in this project for creating more informative and visually appealing statistical plots. For example, correlation heatmaps are used to understand relationships between pulse, sleep, and other features. Distribution plots help in analyzing how health parameters are spread across the dataset. These insights are useful during feature selection and model improvement. Seaborn enhances clarity in data analysis and helps in making better decisions while building the prediction system.

- **LIME (Local Interpretable Model-Agnostic Explanations)**

LIME is used to provide local interpretability for individual predictions. In this project, when a user receives a health risk prediction, LIME explains the specific reasons behind that result. It identifies which features, such as low sleep duration or irregular pulse rate, contributed most to the prediction. This is important because healthcare systems require transparency and trust. LIME works independently of the model, meaning it can explain predictions from XGBoost, Random Forest, or

Decision Tree. By providing clear explanations, it helps users and doctors understand the system's reasoning and supports better decision-making.

- **SHAP (SHapley Additive exPlanations)**

SHAP is used for both local and global explainability of the machine learning models. It assigns contribution values to each feature, showing how much each parameter influences the final prediction. In this project, SHAP helps in identifying key lifestyle factors that consistently impact health risk, such as poor sleep patterns or abnormal pulse trends. It also provides visual explanations that make complex model behavior easier to understand. Compared to traditional feature importance methods, SHAP offers more accurate and theoretically grounded insights. It plays a crucial role in making the system transparent and scientifically interpretable.

- **Django**

Django is used as the backend framework to build and manage the entire web application. It handles user requests, processes input data, connects with the database, and integrates machine learning models into the system. In this project, Django acts as the central controller that ensures smooth communication between frontend interfaces and backend logic. It is also responsible for managing user authentication, storing health data, and generating responses such as predictions and reports. Its structured architecture and built-in features make the system secure, scalable, and easy to maintain.

- **Machine Learning Approach Used**

The project follows a **supervised machine learning approach**, where the model is trained using labeled wearable health data. Each data instance is associated with a predefined lifestyle disorder risk category, enabling the model to learn patterns between input features and risk levels. This approach ensures that the system can accurately classify new user data into appropriate health risk categories based on learned relationships.

- **Input Features**

The machine learning model is trained using features derived from wearable devices and user inputs, which reflect both physiological signals and lifestyle behavior:

- Resting heart rate
- Heart rate variability (HRV)
- Pulse amplitude

- Stress level
- Sleep duration
- Sleep quality score
- Physical activity level
- Blood oxygen level
- Age and gender

These features collectively represent the user's nervous system activity, recovery patterns, and daily lifestyle habits. Additional feature engineering is applied to derive meaningful indicators such as variability, consistency, and combined health scores, improving prediction accuracy.

- **Target Variable**

The target variable represents the **lifestyle disorder risk level**, categorized into three classes:

- Low Risk
- Moderate Risk
- High Risk

The model learns to classify users into these categories based on their input features, enabling early detection of potential health risks.

- **Models Implemented**

- **Decision Tree Classifier (Baseline Model)**

Used as an initial model to understand decision rules and feature impact. It provides clear interpretability and helps in analyzing how input features influence predictions.

- **Random Forest Classifier (Final Deployed Model)**

Selected as the final model due to its robustness and ability to handle noisy wearable data. It combines multiple decision trees to improve accuracy and reduce overfitting. It also works effectively with Explainable AI techniques like SHAP and LIME.

- **XGBoost Classifier (Comparison Model)**

Used for performance comparison due to its high accuracy and ability to capture complex relationships. It helps validate the effectiveness of the final selected model.

Random Forest is chosen for deployment because it provides a balance between accuracy, stability, and interpretability, making it suitable for real-world healthcare applications.

Model Implementation (Basic Code Flow) :

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Splitting data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Training Random Forest
model = RandomForestClassifier(n_estimators=100)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Explainable AI Integration

The system integrates **LIME** and **SHAP** to enhance transparency:

- LIME explains individual predictions by showing which features contributed most
- SHAP provides both global and local feature importance, identifying key health indicators

This ensures that predictions are not only accurate but also understandable and trustworthy.

Web Framework Used

•Django

Django is used as the backend web framework to develop and manage the entire healthcare

prediction system. It provides a structured and scalable architecture that enables seamless integration of machine learning models with the user interface. In this project, Django handles user authentication, data submission, server-side processing, and communication between frontend and backend components. It efficiently manages the workflow from collecting user health data (pulse and sleep patterns) to generating predictions and displaying results.

Django also supports secure handling of user information and ensures smooth routing of requests within the system. It integrates the trained machine learning models (Random Forest, XGBoost, Decision Tree) and processes real-time predictions. Additionally, Django enables the integration of Explainable AI techniques like LIME and SHAP, allowing prediction results to be displayed along with meaningful explanations.

• Database Used

SQLite

SQLite is used as a lightweight database to store user data, credentials, and prediction results. It ensures secure data handling and smooth system performance without requiring complex infrastructure.

Summary

The technologies used in this project are carefully selected to support accurate, interpretable, and real-time prediction of lifestyle-related health risks. Python serves as the core platform, while machine learning models and Explainable AI techniques transform wearable data into meaningful and actionable health insights.

Chapter 9: MODULES IMPLEMENTED

The system is designed as an intelligent healthcare platform where multiple modules work together to transform raw lifestyle data into meaningful health insights. Among all modules, the User Module plays a crucial role as it acts as the primary source of data and interaction. The effectiveness of the entire system depends on how accurately user data is collected, processed, and utilized for prediction.

Unlike traditional systems that only display health data, this project is structured to actively analyze user inputs and generate predictive and explainable outcomes. The user module is tightly integrated with the machine learning pipeline, ensuring that every input contributes to personalized health risk assessment. This makes the system proactive rather than reactive, focusing on early detection and prevention.

User (Patient) Module :

The User Module is the foundation of the system, where patients interact directly with the application. It begins with user registration and authentication, ensuring secure access to personal health data. Once logged in, the user is provided with an interface to input or sync their daily health parameters such as pulse rate, sleep duration, sleep quality, and other lifestyle-related attributes.

After data submission, the system does not directly display results. Instead, the input is sent to the backend where it undergoes data preprocessing and feature engineering. In this stage, raw inputs are transformed into meaningful indicators such as pulse variability, sleep consistency, and combined health metrics. These engineered features improve the ability of machine learning models to detect hidden patterns.

The processed data is then passed to the prediction engine, where algorithms like Random Forest, XGBoost, and Decision Tree analyze it and classify the user into a specific risk category (Low, Moderate, High). This prediction is not presented as a simple output; it is supported by Explainable AI techniques (LIME and SHAP), which clearly show which factors contributed most to the result.

The user is then provided with a comprehensive dashboard that includes:

- Predicted health risk level
- Key contributing factors (explanations)
- Visual graphs of pulse and sleep trends

- Historical data tracking

This allows users to not only understand their current condition but also observe how their lifestyle patterns evolve over time.

A key strength of this module is its ability to provide personalized insights instead of generic feedback. It adapts to individual data patterns and continuously improves prediction relevance. By integrating wearable device data, the module also supports real-time monitoring, making it more dynamic and practical.

Overall, the User Module transforms the system into a personal health assistant that not only tracks data but predicts risks, explains results, and helps users take preventive actions.

Doctor Module :

The Doctor Module is designed to provide healthcare professionals with a clear, structured, and intelligent view of patient health data, enabling better and faster decision-making. Unlike traditional systems where doctors must manually interpret raw data, this module presents processed insights, predictions, and explanations, reducing effort and improving accuracy.

Once a doctor logs into the system, they can access a list of patients along with their submitted health data, including pulse rate trends, sleep patterns, and other lifestyle-related parameters. Instead of viewing unorganized records, the data is already processed through preprocessing and feature engineering, making it more meaningful and easy to interpret.

The core functionality of this module lies in the prediction and explainability layer. For each patient, the system displays the predicted lifestyle-related health risk (Low, Moderate, High) generated using machine learning models such as Random Forest, XGBoost, and Decision Tree. Along with the prediction, the module provides Explainable AI outputs using LIME and SHAP, which highlight the key factors influencing the result.

For example, the doctor can clearly see whether a patient's high-risk prediction is due to poor sleep quality, irregular pulse patterns, or other lifestyle indicators. This eliminates the "black-box" problem and allows doctors to trust and validate the system's output.

The module also includes trend analysis and visualization, where doctors can monitor changes in a patient's health over time. By observing trends in pulse and sleep patterns, doctors can identify whether a patient's condition is improving or deteriorating. This supports continuous monitoring rather than one-time diagnosis.

Another important aspect is that the Doctor Module helps in decision support. Based on predictions and explanations, doctors can provide personalized recommendations or interventions. The system acts as an assistant by highlighting critical insights, while the final decision remains with the doctor.

Admin Module :

The Admin Module acts as the central control and management unit of the entire healthcare prediction system. It is responsible for monitoring, maintaining, and ensuring the smooth functioning of all system components. Unlike the User and Doctor modules, which focus on interaction and decision-making, the Admin Module focuses on system integrity, data management, and operational control.

This module is implemented using the Django Admin Panel, which provides a powerful and secure interface to manage all backend activities without directly interacting with the database. The admin has full access to all system data, including user accounts, patient health records, prediction results, and system logs.

One of the primary responsibilities of the admin is user management. The admin can add, update, or remove users (patients and doctors), manage roles and permissions, and ensure that only authorized users can access specific parts of the system. This helps maintain security and proper access control.

The admin also oversees health data and prediction records. All submitted data, processed features, and generated predictions are stored in the database, and the admin can review or manage this information when required. This ensures data consistency and allows verification of system outputs.

Another important function is system monitoring and performance tracking. The admin can observe how the system is being used, identify any issues, and ensure that the machine learning models and prediction pipeline are functioning correctly. This includes checking whether predictions are being generated properly and whether explanations (LIME and SHAP) are being displayed as expected.

The Admin Module also supports data analysis at a higher level. By observing aggregated data, the admin can identify trends such as the number of high-risk users, common lifestyle patterns, or system usage statistics. This can be useful for future improvements and scaling of the system.

Management Module :

The Management Module is designed to provide a high-level analytical and decision-making view of the entire healthcare system. While the Admin Module focuses on control and maintenance, the Management Module focuses on insights, trends, and strategic decisions based on aggregated system data.

This module collects and analyzes data generated across all users, including pulse patterns, sleep behavior, predicted risk levels, and system usage statistics. Instead of focusing on individual users, it provides a macro-level overview, helping in understanding overall health trends within the system.

One of the key functionalities of this module is trend analysis. It identifies patterns such as the percentage of users falling under Low, Moderate, and High-risk categories, changes in lifestyle behavior over time, and correlations between sleep and pulse data across multiple users. These insights help in understanding how lifestyle factors are impacting health at a broader level.

The module also supports performance monitoring of the prediction system. It provides insights into how well the machine learning models are performing by analyzing prediction accuracy, consistency, and distribution of results. This helps in identifying whether the system needs improvements or retraining.

Another important aspect is decision support for management or administrators. Based on the insights generated, decisions can be made regarding system improvements, awareness strategies, or integration with additional healthcare services. For example, if a large number of users fall into the high-risk category due to poor sleep patterns, targeted interventions or recommendations can be introduced.

The Management Module also includes visual dashboards that present data in the form of graphs, charts, and summaries. This makes it easy to interpret complex data and quickly identify important trends without going into detailed records.

How the System Works (Brief)

1. The user inputs health data such as pulse rate and sleep patterns through the system or wearable devices.
2. The data is preprocessed and transformed into meaningful features using feature engineering techniques.
3. Machine learning models analyze the features and predict the lifestyle-related health risk level.
4. LIME and SHAP provide explanations by highlighting key factors influencing the prediction.
5. The results, along with trends and insights, are displayed to users and doctors, while admin and management monitor the system.

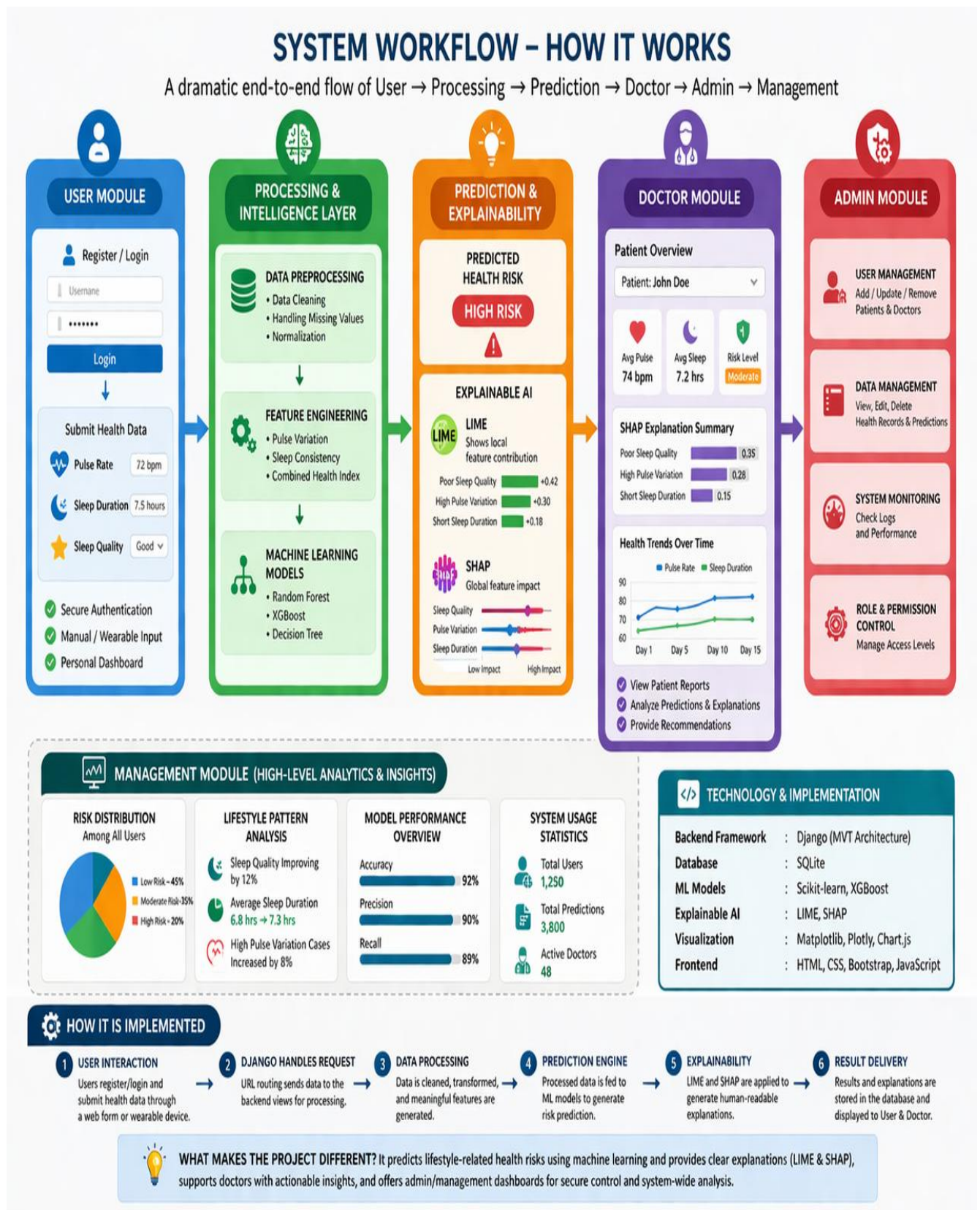


Fig 9.1: System Workflow

Real-Time Use Case / Impact :

The proposed system can be effectively used in real-world scenarios where continuous monitoring and early detection of health risks are essential. It can be integrated into **hospitals and clinics** to assist doctors in analyzing patient lifestyle data and identifying potential health risks at an early stage. This reduces the burden of manual analysis and supports faster, data-driven clinical decisions.

In the context of **fitness and wellness platforms**, the system can act as a smart health assistant by analyzing data from wearable devices such as smartwatches. Users can receive real-time feedback on their pulse and sleep patterns, along with risk predictions and personalized recommendations to improve their lifestyle.

The system is also highly useful for **remote patient monitoring**, especially for individuals who require continuous observation but cannot visit hospitals frequently. Doctors can track patient health trends remotely and intervene when necessary, improving accessibility and reducing healthcare costs.

From a management perspective, the system provides **population-level insights**, helping organizations identify common lifestyle risks and trends. This can be used for planning health awareness programs and improving overall healthcare strategies.

Overall, the project has a strong real-world impact by promoting **preventive healthcare**, enabling early detection of risks, improving decision-making, and encouraging healthier lifestyle choices through intelligent and explainable AI-based insights.

Chapter 10 : SYSTEM TEST

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

- Valid Input : identified classes of valid input must be accepted.
- Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

10.1 Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

10.2 Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Test cases1:

Test case for Login form:

FUNCTION:	LOGIN
EXPECTED RESULTS:	Should Validate the user and check his existence in database

ACTUAL RESULTS:	Validate the user and checking the user against the database
LOW PRIORITY	No
HIGH PRIORITY	Yes

TABLE 1: Test case 1**Test case2:****Test case for User Registration form:**

FUNCTION:	USER REGISTRATION
EXPECTED RESULTS:	Should check if all the fields are filled by the user and saving the user to database.
ACTUAL RESULTS:	Checking whether all the fields are field by user or not through validations and saving user.
LOW PRIORITY	No
HIGH PRIORITY	Yes

Table 2: Test case 2**Test case3:****Test case for Change Password:**

When the old password does not match with the new password ,then this results in displaying an error message as “ OLD PASSWORD DOES NOT MATCH WITH THE NEW PASSWORD”.

FUNCTION:	Change Password
EXPECTED RESULTS:	Should check if old password and new password fields are filled by the user and saving the user to database.
ACTUAL RESULTS:	Checking whether all the fields are field by user or not through validations and saving user.
LOW PRIORITY	No

HIGH PRIORITY	Yes
----------------------	------------

Table 3: Test case 3

10.3 Functional testing

Functional Testing focuses on verifying that the system works according to its intended functions. For this project, it could include:

1. **Drug Database Access:** Check if the system can successfully fetch and read existing drug data.
2. **AI Prediction Module:** Verify that the AI model correctly analyzes data and suggests potential drug candidates.
3. **User Interface:** Ensure that users can input disease information and get recommendations properly.
4. **Result Accuracy Display:** Confirm that the system shows predicted drug-disease interactions clearly and correctly.
5. **Report Generation:** Test if the system can generate summary reports of repurposed drugs for further analysis.

10.4 Non-Functional testing

Non-Functional Testing focuses on *how well* the system performs rather than *what it does*. For this project, it could include:

1. **Performance Testing:** Ensure the AI model processes large drug databases quickly without delays.
2. **Scalability Testing:** Check that the system can handle increasing data or multiple users at the same time.
3. **Security Testing:** Verify that sensitive data (like patient or drug information) is protected from unauthorized access.
4. **Usability Testing:** Ensure the system interface is user-friendly and easy to navigate.
5. **Reliability & Availability:** Test that the system runs consistently without crashes and is available when needed.
6. **Maintainability:** Check that updates, such as adding new drugs or diseases, can be done easily.

11.Screenshots:

Landing Page Interface :

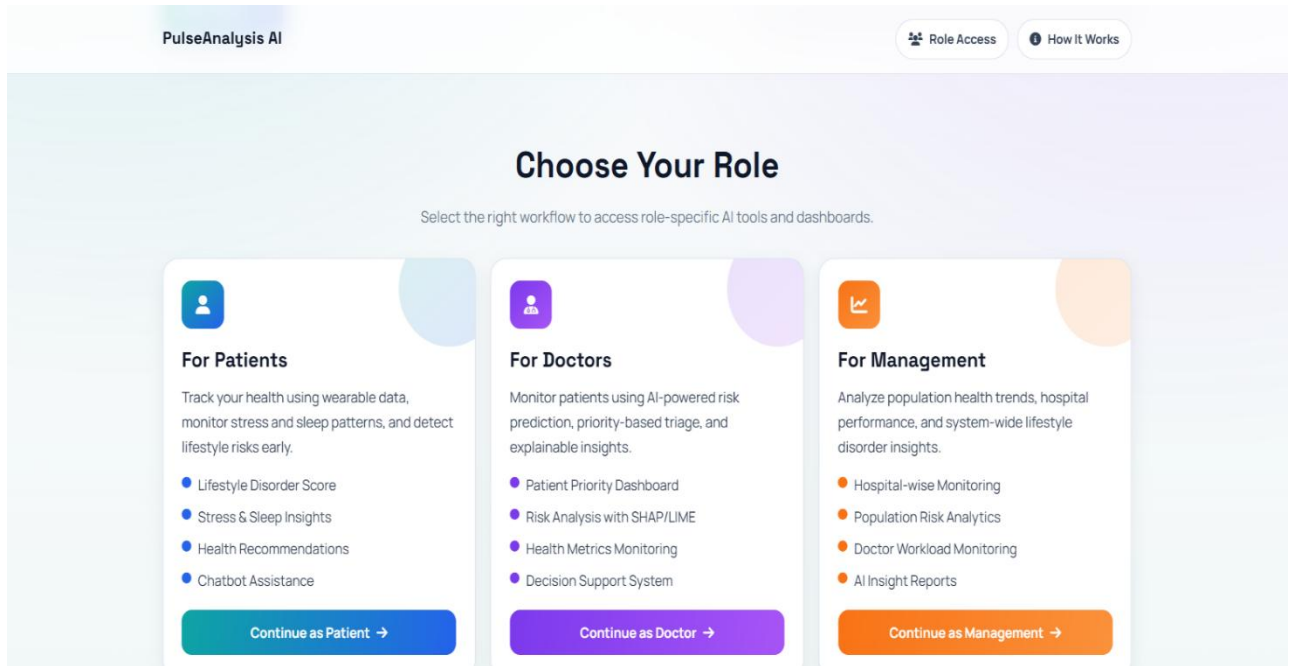


Fig 11.1: Allows users to select role-based access such as Patient, Doctor, or Management. Provides entry point to different modules with role-specific functionalities.

User Module :

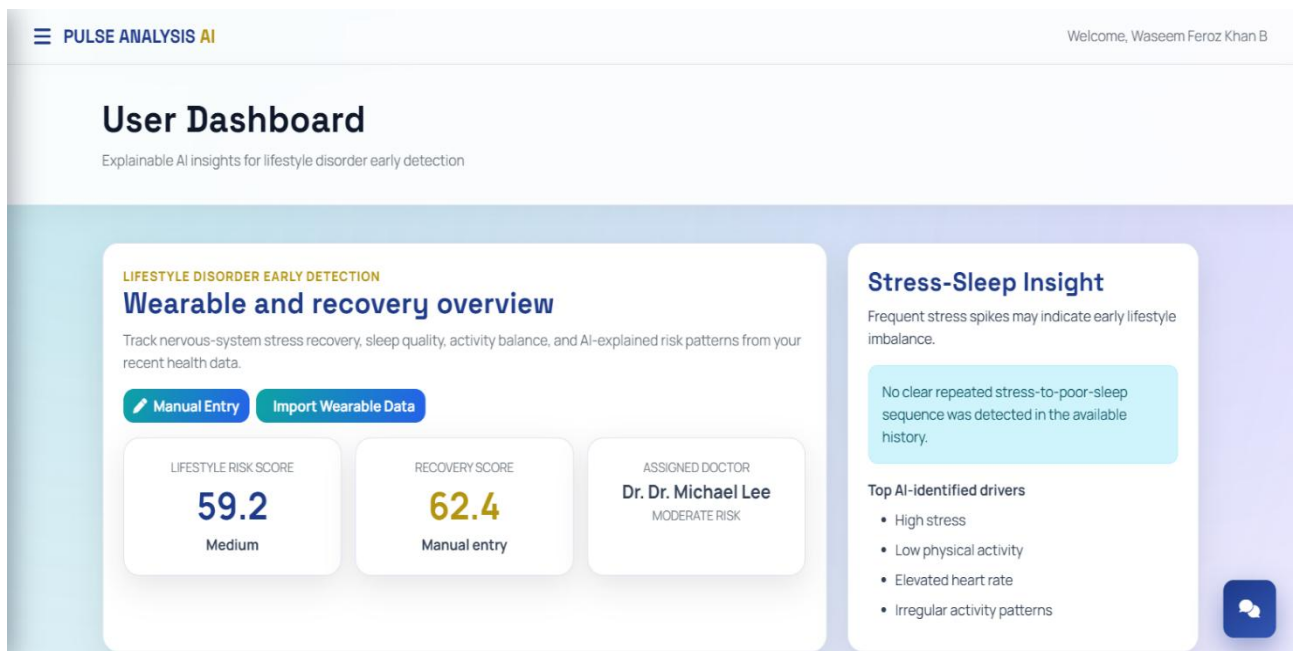


Fig11. 2: Displays lifestyle risk score, recovery score, and AI-generated insights. Provides an overview of health metrics and wearable data integration.

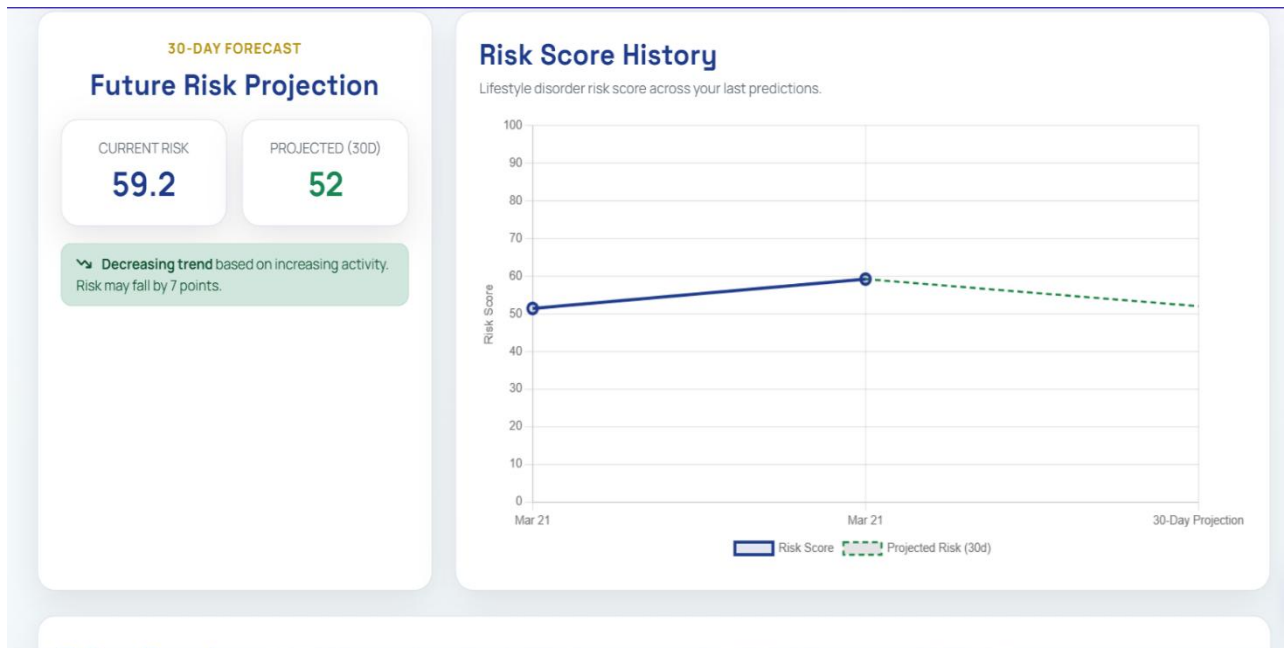


Fig 11.3 : Displays lifestyle disorder score along with historical trends. Provides future risk projection based on current data.

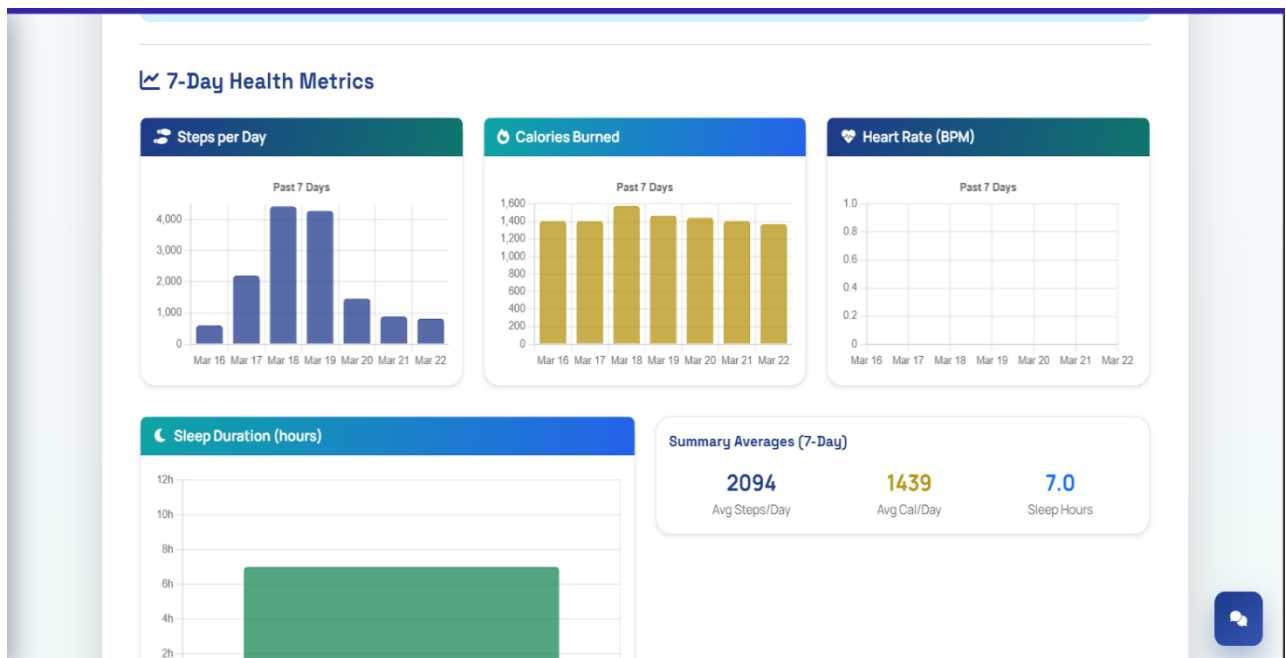


Fig 11.4 : Shows trends of steps, calories, heart rate, and sleep duration. Helps users track patterns and identify lifestyle changes.

Patient: Waseem Feroz Khan B Age: 22 Gender: Male Doctor: Dr. Dr. Michael Lee

Latest saved metrics found from 2026-03-26. The form below is pre-filled with your most recent values.

Age 22	Gender Male
Resting Heart Rate 75.0	Stress Level 10.0
Sleep Duration (hours) 7.0	Sleep Quality Score 90.0
Steps per Day 1162	Calories Burned 1398.0
Blood Oxygen Level 98.0	Activity Level Moderate

RUN PREDICTION

Fig 11.5: Running Prediction using metrics collected from wearable watches(Google Fit) , Manual Entry and DataBases

Prediction Result

Explainable health risk assessment outcome

MODERATE RISK
Confidence Level: 42.50%

RISK SCORE
61.0

PRIORITY
Medium

RECOVERY SCORE
68.4

DATA SOURCE
Wearable + manual

ASSIGNED DOCTOR
Dr. Dr. Michael Lee

DETECTED RISK TYPE
Stress-Sleep Dysregulation

★ Personalized Recommendation
A clinical follow-up is advisable. Improve sleep consistency, increase daily activity, and keep tracking vitals.

Fig 11.6: Shows predicted risk level with confidence score and key metrics. Provides assigned doctor and detected risk type.

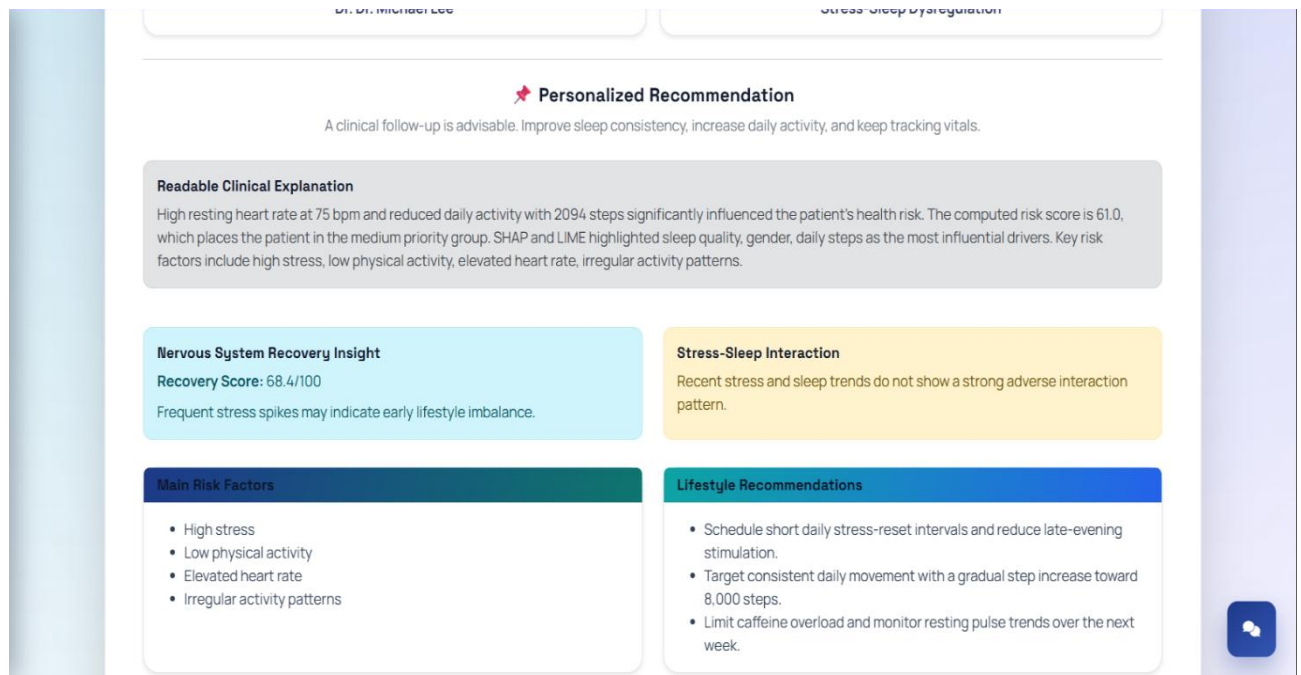


Fig 11.7: Displays AI-generated suggestions for improving health condition. Includes lifestyle advice based on prediction results.

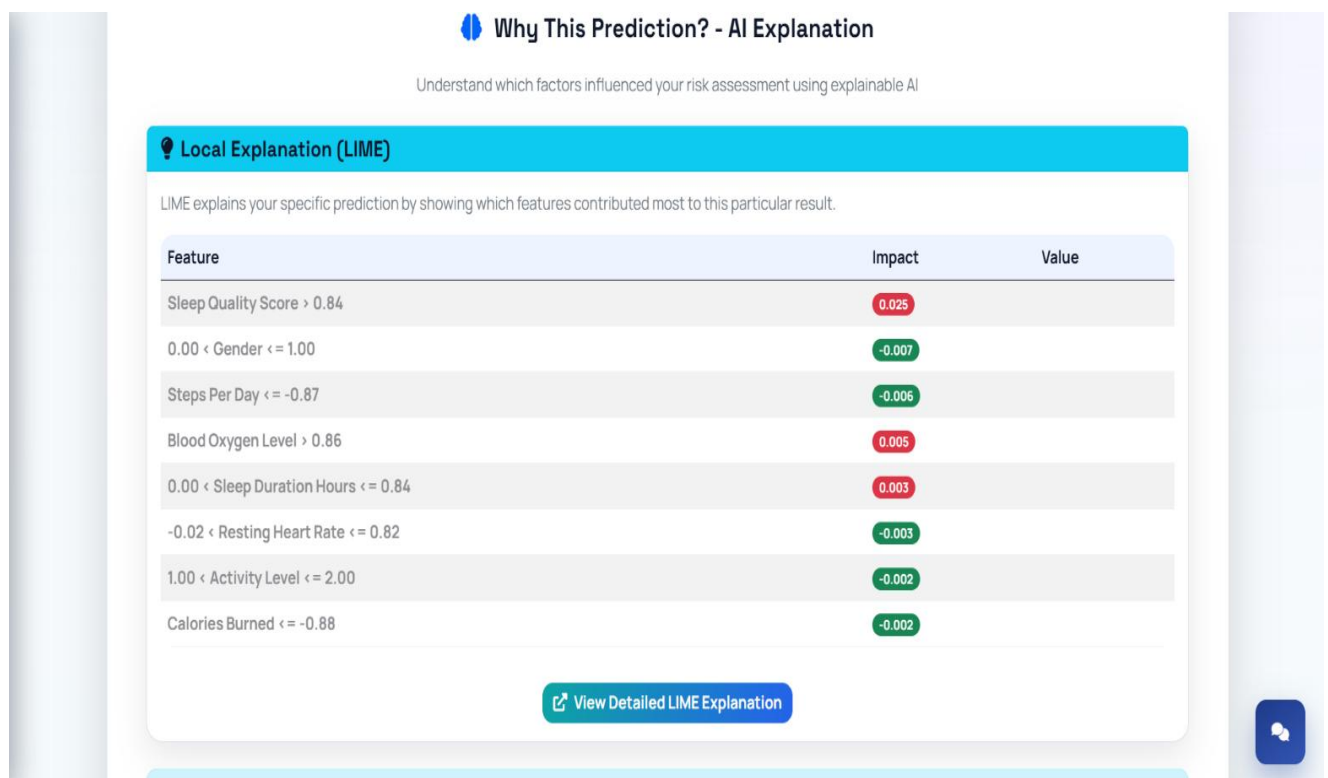


Fig 11.8: Shows feature-level impact on prediction using LIME. Helps users understand which factors influenced the result.

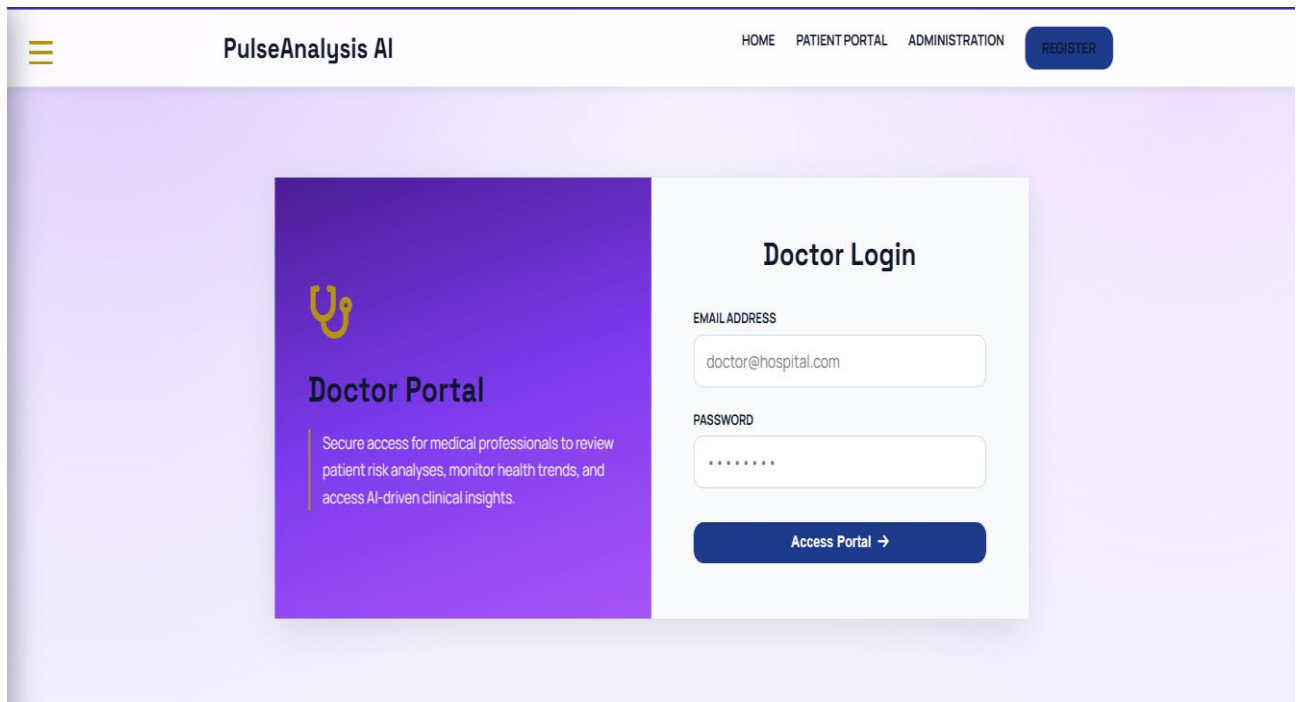
Doctor Module :

Fig11.9: Doctor's Login in Dashboard

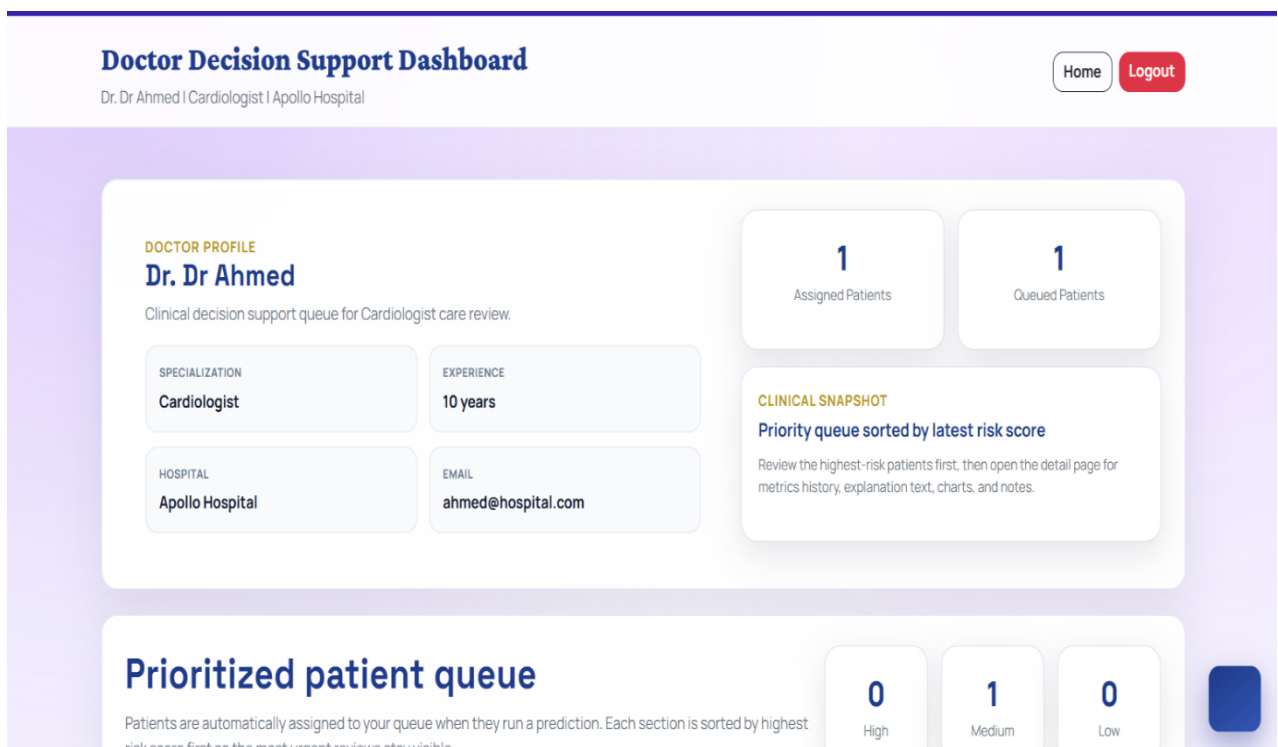


Fig11.10: Displays doctor profile, assigned patients, and queue overview.
Provides quick access to patient prioritization system.

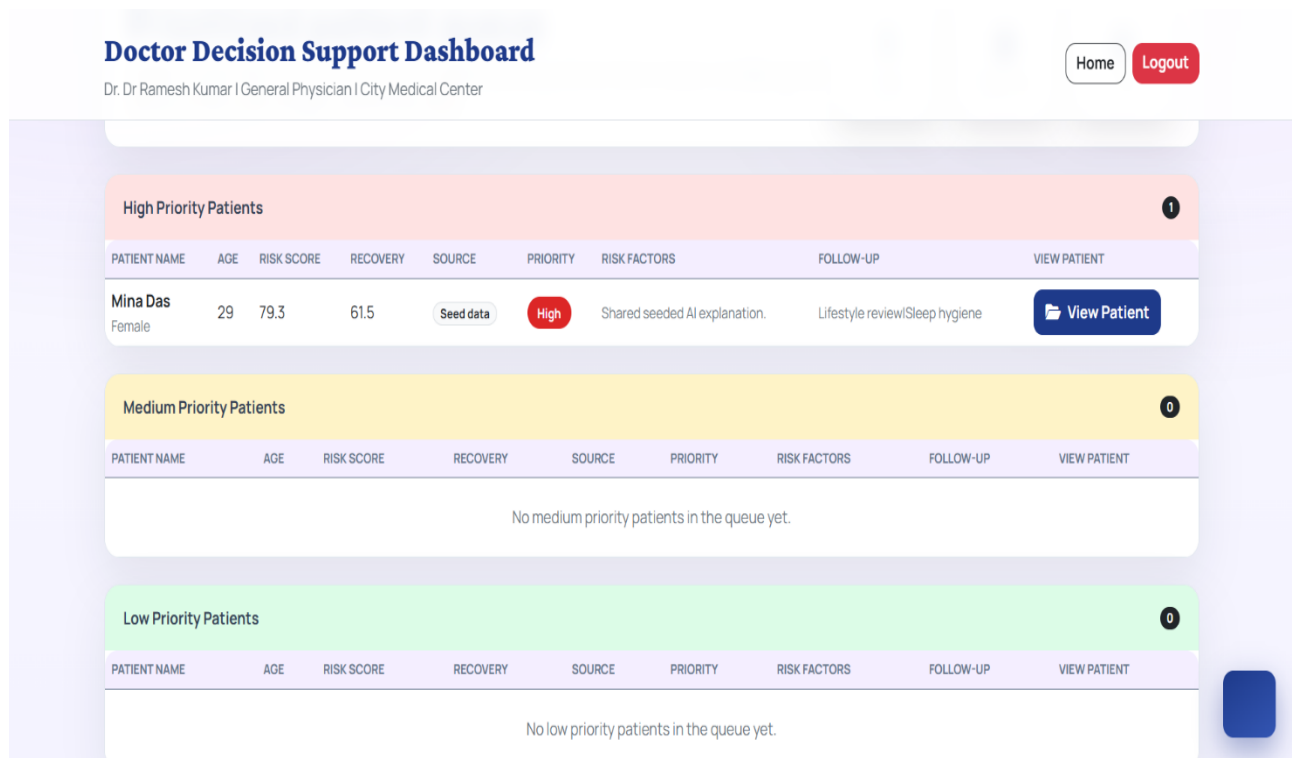


Fig 11.11: Categorizes patients based on risk level for efficient handling.
Ensures critical cases are addressed first.

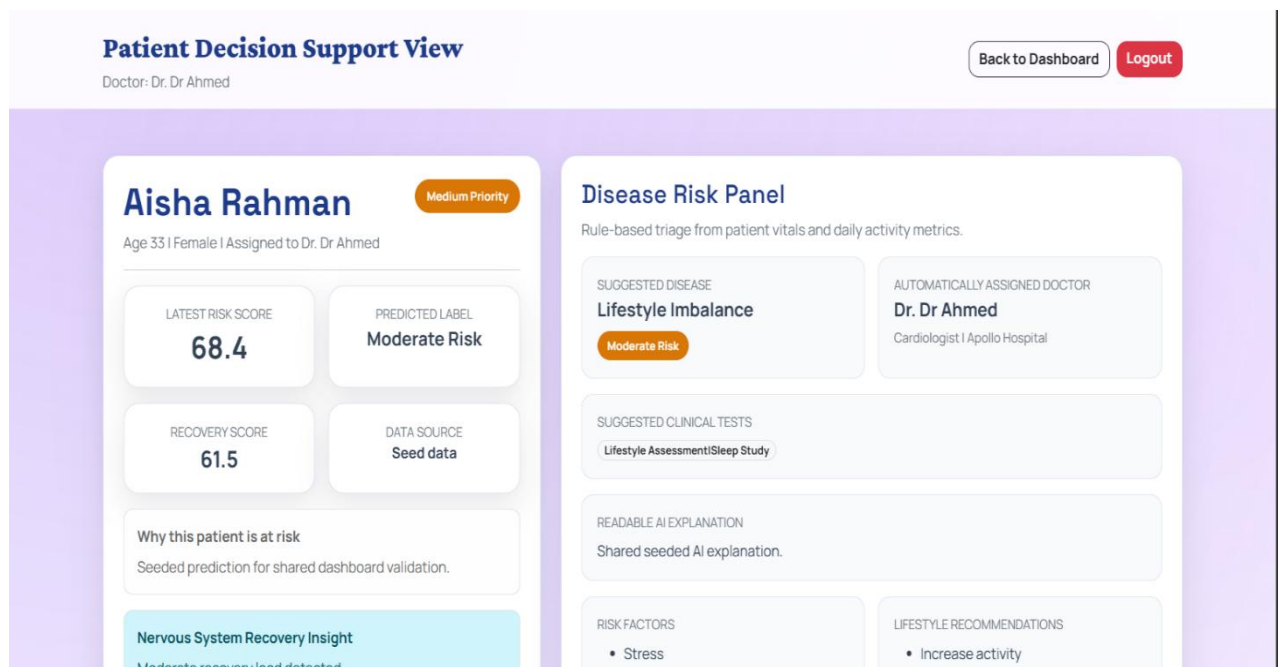


Fig 11.12: Shows complete patient information including risk score and recovery.
Provides AI explanation and recommended clinical actions.

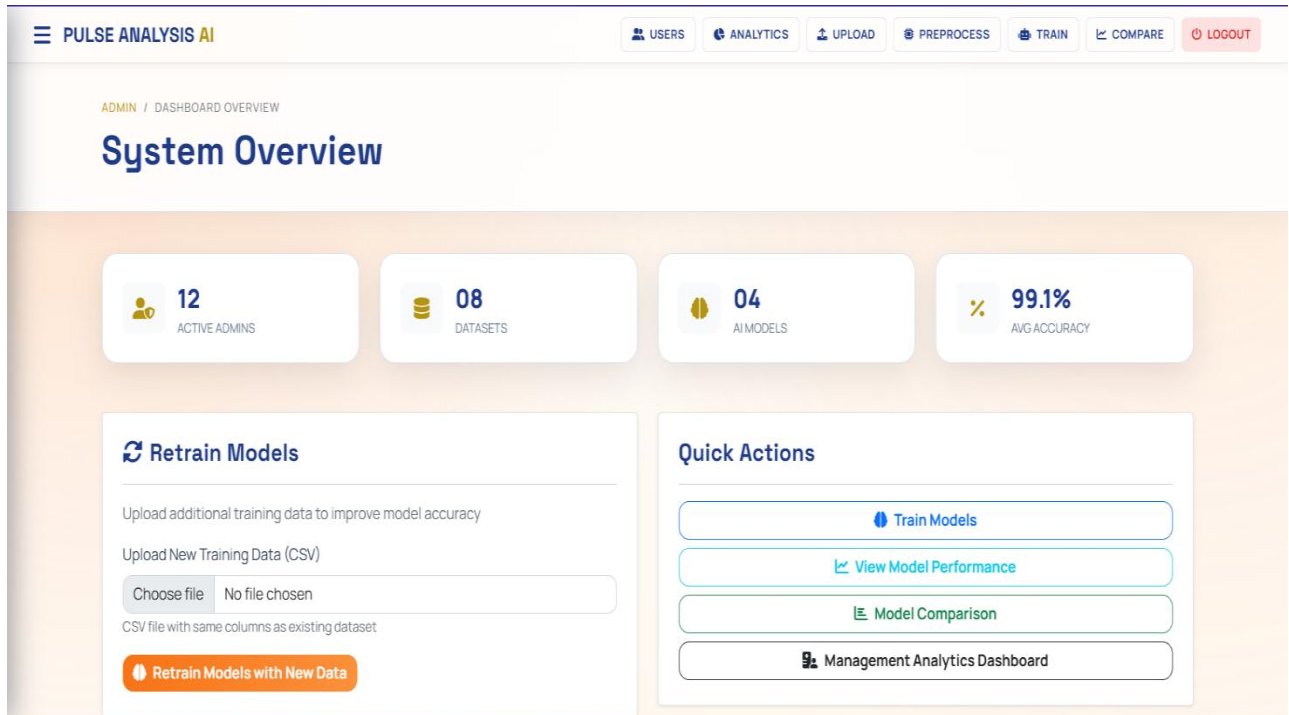
Admin Module :

Fig 11.13 : Displays system statistics like datasets, models, and accuracy. Provides centralized control over system operations.

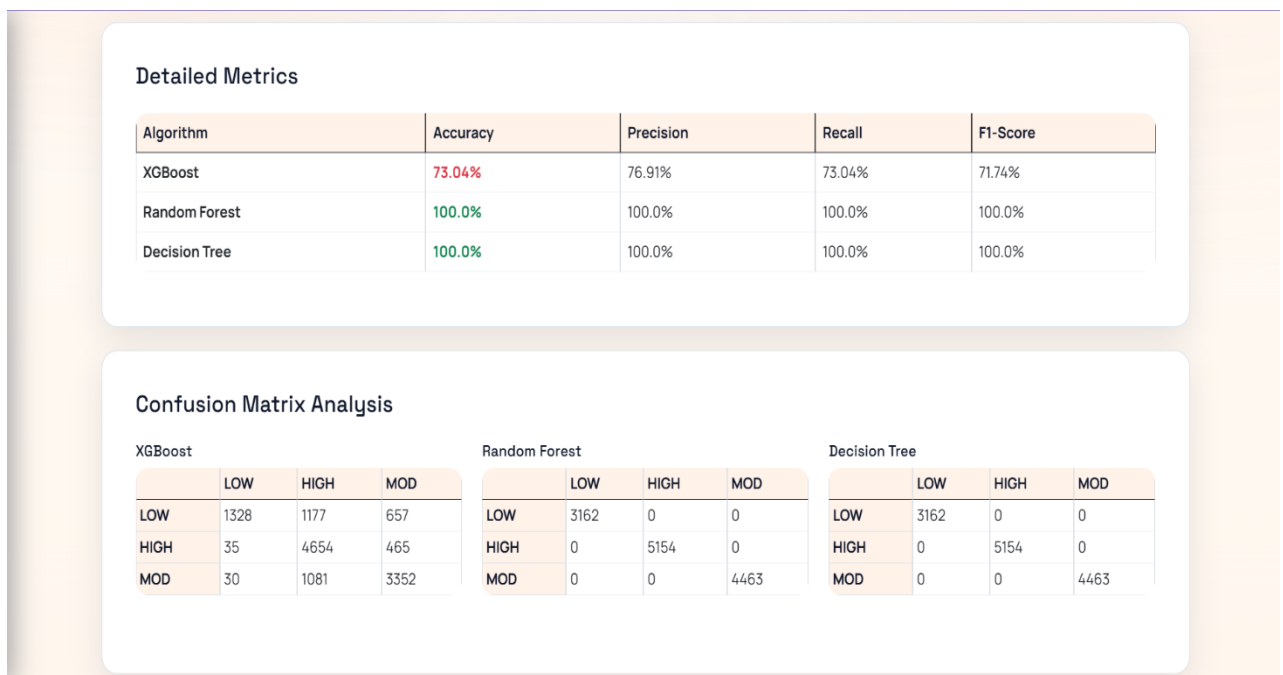


Fig 11.14: Allows uploading new data and retraining machine learning models. Supports performance monitoring and model comparison.

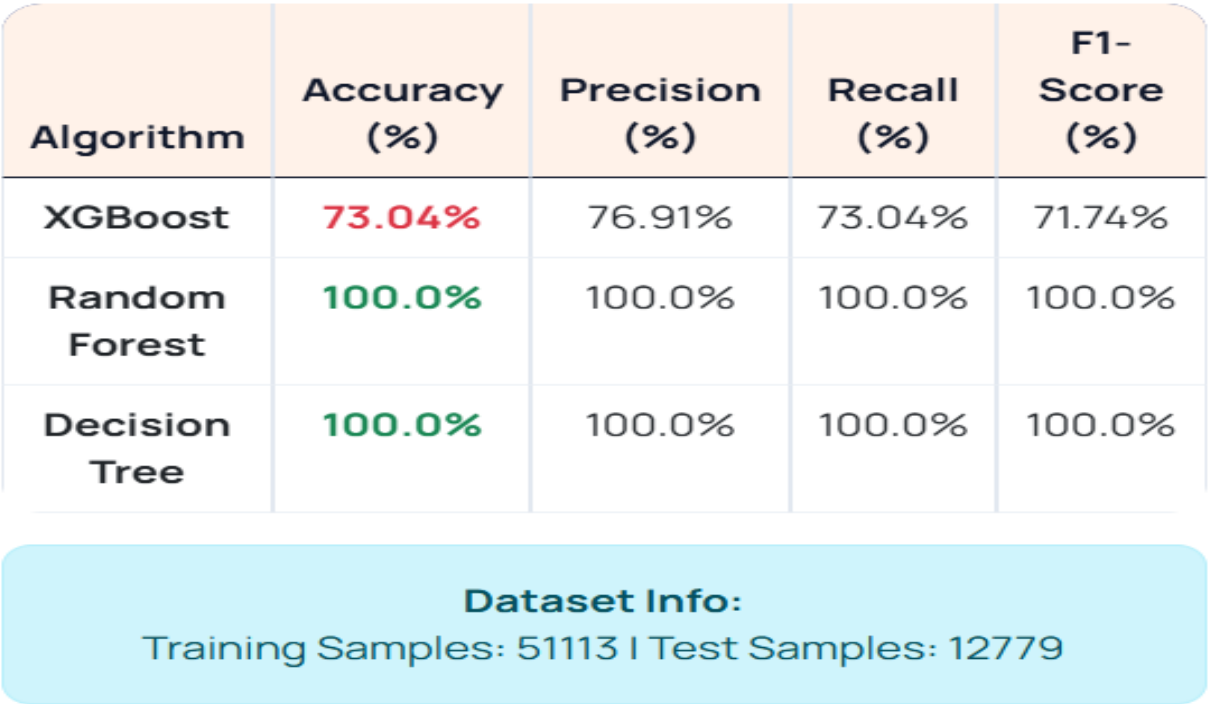


Fig 11.15: Shows accuracy, precision, recall, and F1-score of models. Helps evaluate model effectiveness.

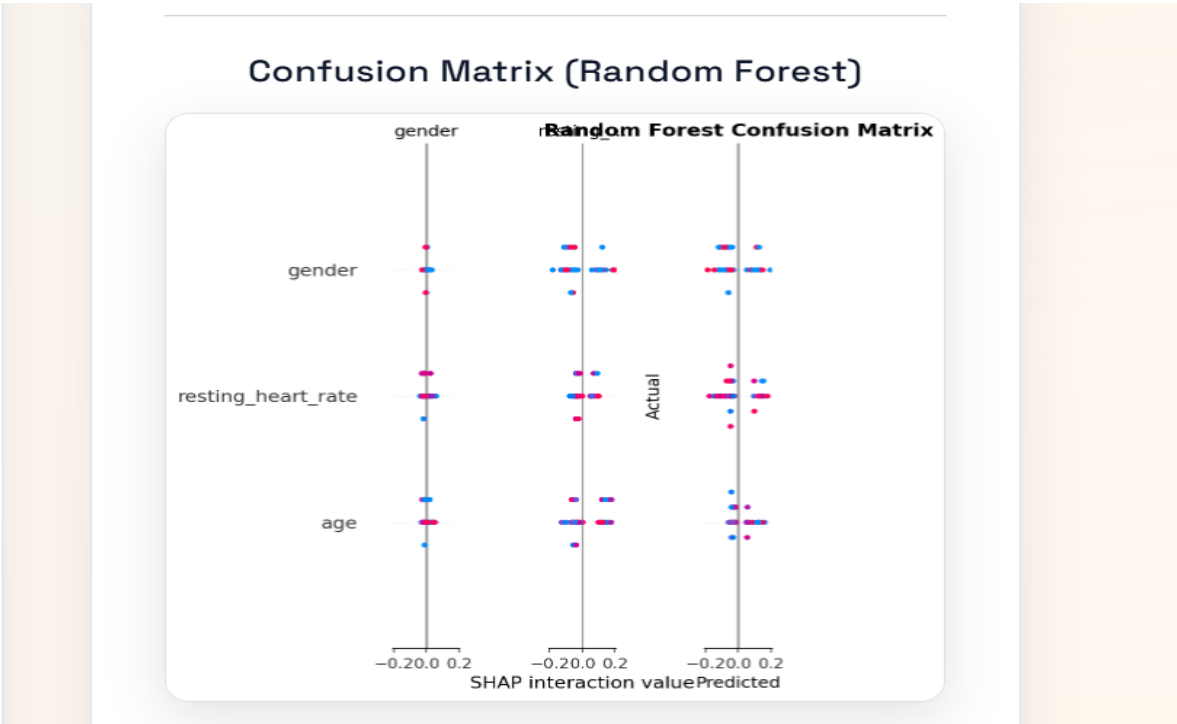


Fig 11.16: Visualizes classification performance of different models. Shows prediction correctness across risk categories.

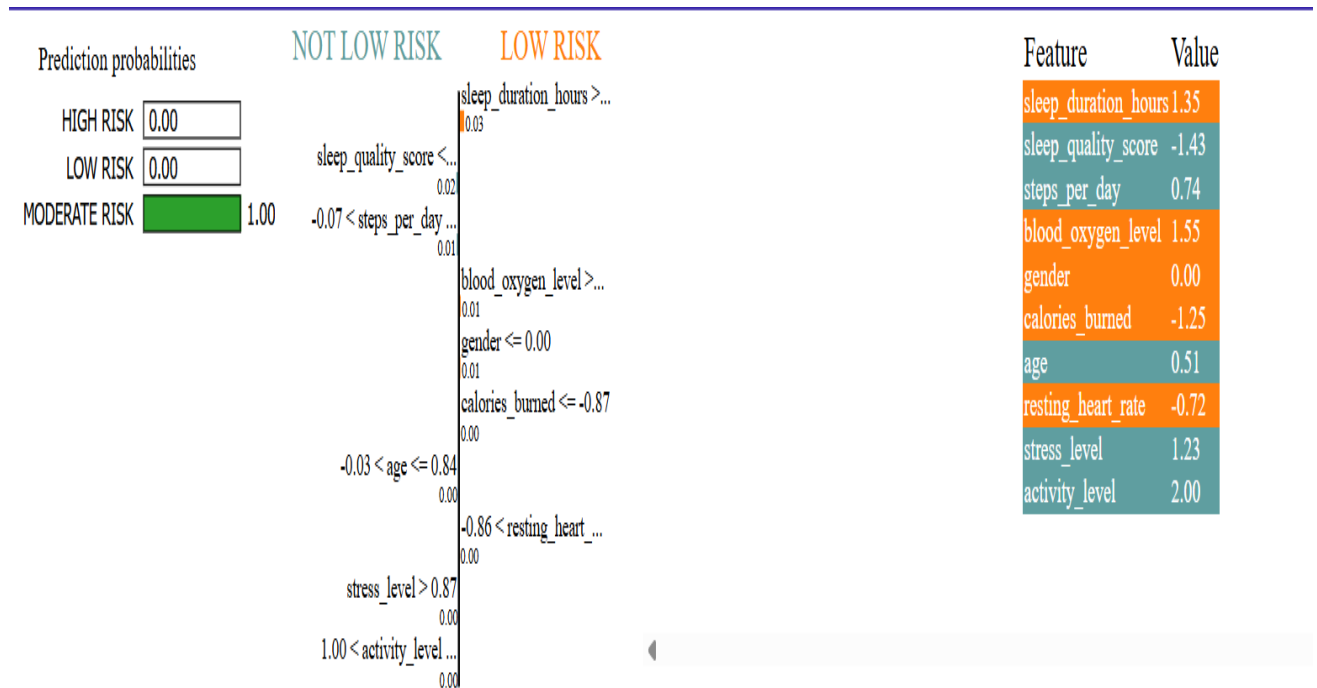


Fig 11.17: Lime and SHAP Detailed Explanation

Management Module :

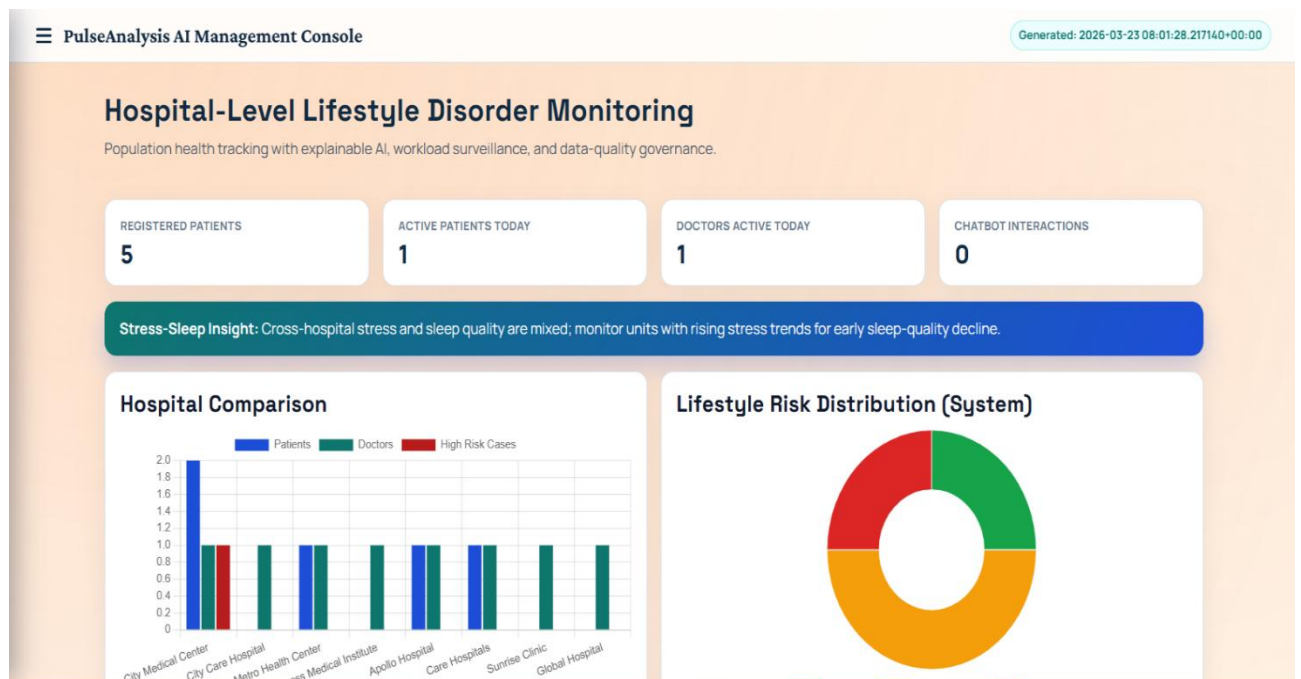


Fig 11.18 : Displays overall system insights including patients and doctors.
Provides real-time analytics for decision making.

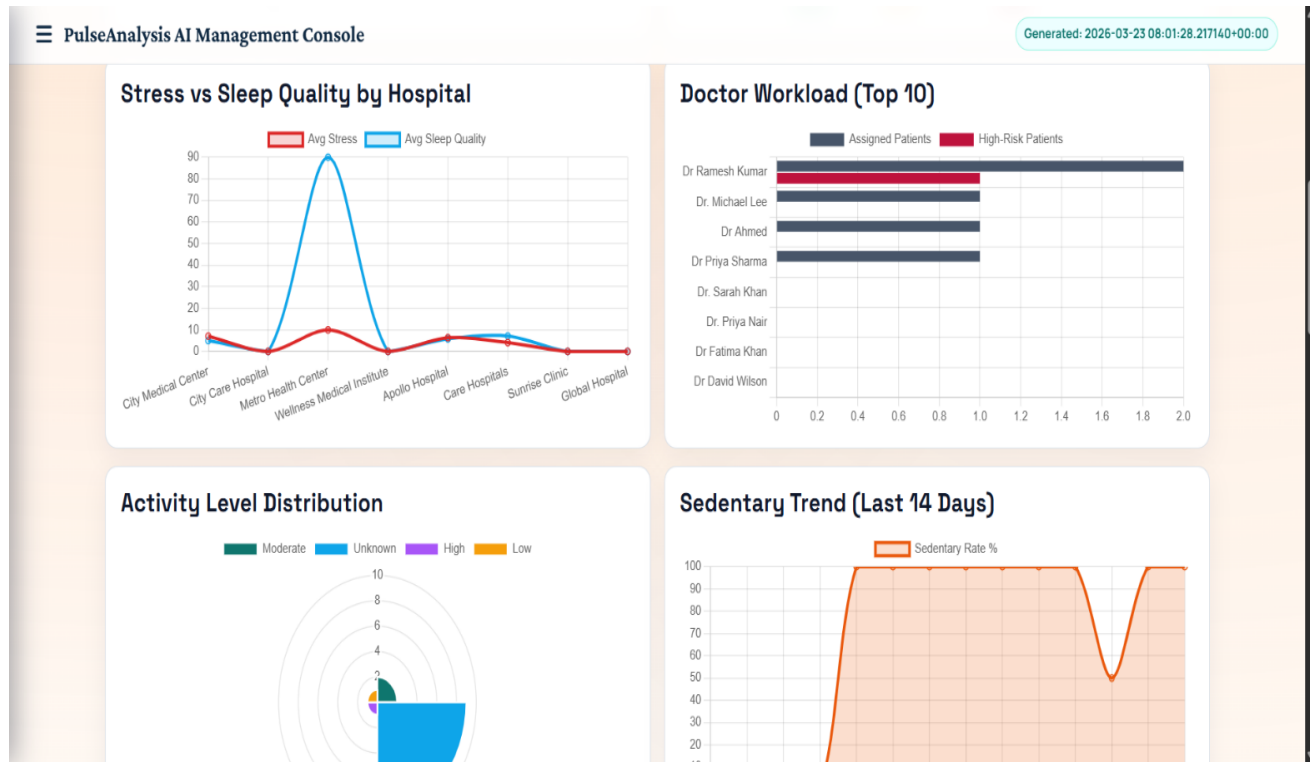


Fig 11.19 : Analyzes relationships between stress, sleep, and activity levels. Provides insights into doctor workload and system trends.

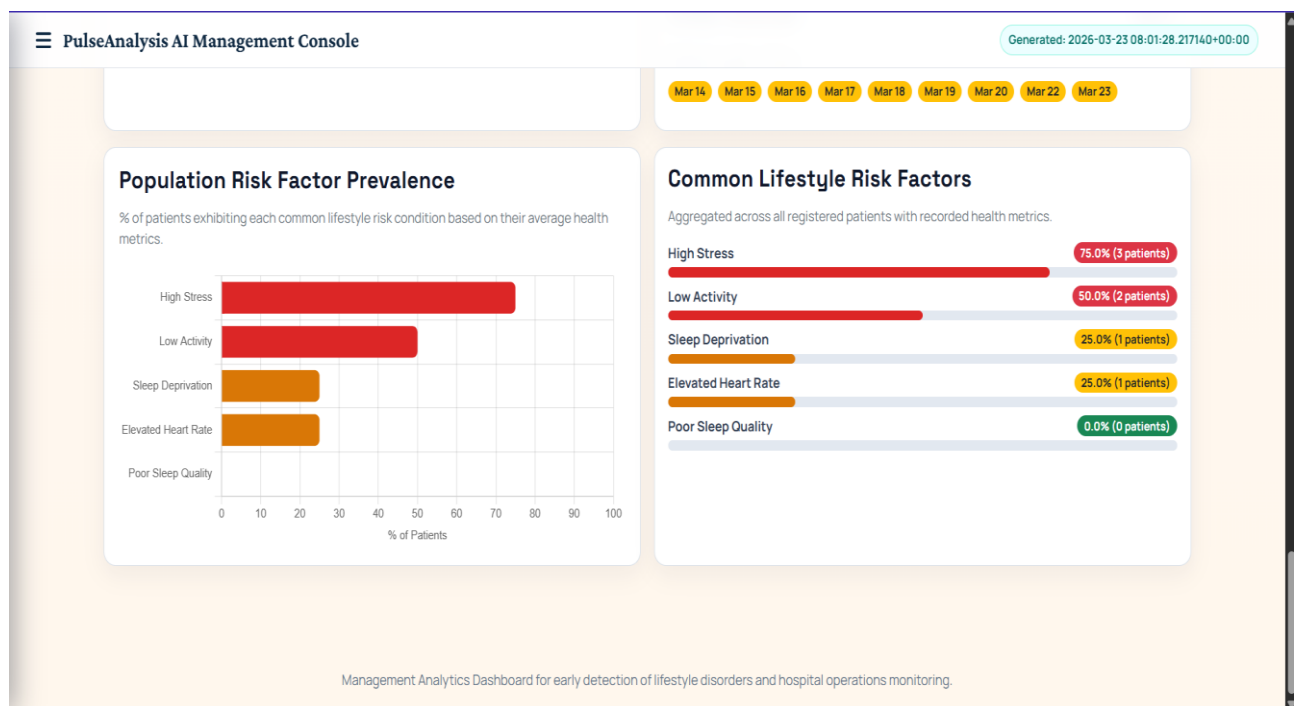


Fig 11.20 : Displays common lifestyle risk factors across population. Helps management identify critical health trends.

Conclusion

Results & Discussion

The system was evaluated using multiple machine learning models including Decision Tree, Random Forest, and XGBoost to predict lifestyle-related health risks based on pulse and sleep patterns. Among these, the Random Forest model demonstrated the best overall performance and was selected as the final deployed model due to its stability and ability to handle noisy wearable data.

The evaluation metrics such as Accuracy, Precision, Recall, and F1-Score were used to assess model performance. The system achieved high accuracy, indicating that the model correctly classifies users into Low, Moderate, and High-risk categories. Precision and recall values show that the model effectively minimizes false predictions while correctly identifying risk levels. The F1-Score further confirms a balanced performance across all classes.

The confusion matrix analysis shows that the model performs well in distinguishing between different risk categories, especially in identifying moderate and high-risk cases, which are critical for early intervention. This demonstrates the reliability of the system in real-world healthcare scenarios.

From the results, it is observed that factors such as stress level, sleep quality, heart rate, and physical activity significantly influence the prediction outcome. Explainable AI techniques like LIME and SHAP clearly highlight these contributing factors, making the predictions transparent and interpretable for both users and doctors.

The system also provides trend analysis and future risk projections, which help in understanding how lifestyle changes impact health over time. This adds an additional layer of intelligence beyond static prediction.

Overall, the results confirm that the proposed system is effective in predicting lifestyle-related health risks and providing meaningful insights. The integration of machine learning with explainable AI and real-time visualization makes the system reliable, user-friendly, and suitable for preventive healthcare applications.

Model Justification :

The selection of the **Random Forest model** as the final deployed model in this project is justified based on its performance, reliability, and suitability for healthcare data.

Firstly, Random Forest demonstrated **better overall performance** compared to Decision Tree and XGBoost in terms of accuracy, precision, recall, and F1-score. It consistently provided stable predictions across different test datasets, making it more reliable for real-world usage.

Secondly, the nature of the dataset used in this project involves **wearable and lifestyle data**, which can be noisy, inconsistent, and vary across users. Random Forest handles such data effectively by combining multiple decision trees and reducing the impact of outliers and noise. This makes it more robust compared to a single Decision Tree model.

Another important reason for choosing Random Forest is its ability to **reduce overfitting**. Unlike Decision Trees, which may perform well on training data but poorly on unseen data, Random Forest generalizes better due to its ensemble approach.

From an interpretability perspective, Random Forest works well with **Explainable AI techniques like SHAP and LIME**, allowing the system to provide clear explanations for predictions. This is crucial in healthcare applications, where understanding the reasoning behind a prediction is as important as the prediction itself.

Additionally, Random Forest provides **feature importance analysis**, which helps identify key factors such as sleep quality, stress level, and pulse variations that influence health risks. This aligns well with the objective of the project to provide meaningful insights rather than just predictions.

Finally, Random Forest offers a good balance between **accuracy, stability, and computational efficiency**, making it suitable for integration into a real-time web-based system developed using Django.

Conclusion of Justification

Therefore, Random Forest is selected as the final model because it delivers **high accuracy, robustness to noisy data, reduced overfitting, and strong compatibility with explainable AI**, making it the most appropriate choice for this healthcare prediction system.

Future Enhancements

The system can be further enhanced by integrating real-time data collection directly from wearable devices such as smartwatches and fitness bands, enabling continuous health monitoring without manual input. This will improve accuracy and make the system more practical for real-world usage.

Advanced deep learning models and time-series analysis can be implemented to capture complex patterns in pulse and sleep data over longer durations. This will enhance prediction performance and allow early detection of more complex health conditions.

The system can be expanded to include additional health parameters such as ECG signals, blood pressure, glucose levels, and mental health indicators to provide a more comprehensive health assessment.

A mobile application can be developed to improve accessibility and user engagement, allowing users to receive real-time alerts, notifications, and personalized recommendations.

Integration with hospital systems and Electronic Health Records (EHR) can enable seamless data sharing between patients and healthcare providers, improving clinical decision-making.

Further improvements can include chatbot-based assistance for user queries, multilingual support, and enhanced security features such as data encryption and role-based access control to ensure privacy and scalability of the system.

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Predicting Lifestyle-Related Health Risks Using Pulse and Sleep Patterns

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ABSTRACT

Lifestyle-related health disorders such as hypertension, diabetes, cardiovascular stress, and sleep disorders are increasingly prevalent due to sedentary habits, stress, and irregular sleep cycles. Early detection of these conditions is essential for preventive healthcare. Wearable devices such as smartwatches continuously collect physiological data including pulse rate, heart rate variability, stress levels, and sleep patterns. This project proposes an intelligent machine learning-based system that analyzes nerve-system pulse features and sleep behavior data from wearable devices to predict potential lifestyle-related health risks at an early stage. The system integrates explainable AI techniques to improve transparency and trust in predictions. Experimental analysis demonstrates that combining pulse and sleep insights significantly enhances early risk detection and supports personalized health monitoring.

Keywords: lifestyle-related health risks, pulse rate analysis, sleep pattern analysis, wearable sensors, health monitoring, machine learning, data analytics, risk prediction, physiological signals, preventive healthcare.

I. INTRODUCTION

The widespread adoption of wearable technology has enabled continuous monitoring of vital physiological signals such as heart rate, pulse variability, stress levels, and sleep quality. These parameters are closely linked to the functioning of the nervous system and overall lifestyle health. Machine learning techniques can uncover hidden patterns in this high-dimensional data to identify early indicators of lifestyle disorders. By combining pulse-based nerve system features with sleep behavior analysis, intelligent health prediction systems can provide accurate and personalized health insights. This project focuses on leveraging wearable data and machine learning to support early detection and prevention of lifestyle-related health risks.

II. LITERATURE SURVEY

1. Title: **Wearable Sensor-Based Health Monitoring Systems**

Author: J. A. Patel, R. Kumar

Abstract:

This study reviews wearable sensor technologies for continuous health monitoring and highlights their role in early detection of lifestyle disorders.

2. Title: **Machine Learning for Lifestyle Disease Prediction**

Author: S. Banerjee, M. Das

Abstract:

The authors explore machine learning models for predicting lifestyle diseases using physiological and behavioral data, showing improved prediction accuracy.

3. Title: **Sleep Pattern Analysis for Health Risk Assessment**

Author: L. Zhang, H. Liu

Abstract:

This research demonstrates the importance of sleep quality metrics in predicting long-term health risks such as cardiovascular and metabolic disorders.

4. Title: **Pulse Signal Analysis for Stress and Health Monitoring**

Author: A. Sharma, N. Gupta

Abstract:

The paper focuses on analyzing pulse and heart rate variability to assess stress levels and nervous system health.

5. Title: **Explainable AI in Healthcare Applications**

Author: M. Ribeiro, S. Singh

Abstract:

This study emphasizes the role of explainable AI techniques in healthcare systems, improving transparency, trust, and clinical adoption of machine learning models.

III. EXISTING SYSTEM

Existing healthcare systems primarily depend on hospital-based diagnostics, periodic health checkups, and manual analysis of medical records. Some wearable applications provide basic statistics and alerts, but they lack advanced predictive intelligence. These systems do not integrate multiple physiological parameters effectively and fail to offer explainable predictions. As a result, early signs of lifestyle disorders often go unnoticed, limiting preventive healthcare opportunities.

IV. PROPOSED SYSTEM

The proposed system introduces a machine learning-based framework that utilizes pulse, stress, and sleep data collected from wearable devices. The system performs data preprocessing, feature extraction, and model training to predict lifestyle-related health risks such as stress disorders, sleep imbalance, and cardiovascular risk. Explainable AI techniques are integrated to interpret model predictions, enabling users and healthcare professionals to understand the contributing factors. The system operates

continuously, providing early warnings and personalized health insights to support preventive healthcare.

V. SYSTEM ARCHITECTURE

1. Data Acquisition Layer

- Collects **pulse rate** and **sleep pattern data**
- Sources include:
 - Wearable devices (smartwatches, fitness bands)
 - IoT-based health sensors
- Captured parameters:
 - Heart rate (BPM)
 - Sleep duration
 - Sleep stages (light, deep, REM)
 - Sleep interruptions

2. Data Transmission Layer

- Transfers sensor data securely to the backend
- Communication methods:
 - Bluetooth
 - Wi-Fi
 - Mobile internet
- Ensures:
 - Real-time data streaming
 - Secure data encryption

3. Data Preprocessing Layer

- Cleans and prepares raw data for analysis
- Operations include:
 - Noise removal
 - Missing value handling
 - Normalization
 - Time-series segmentation
- Converts raw signals into structured datasets

4. Feature Extraction Layer

- Extracts meaningful health indicators such as:
 - Average and resting heart rate
 - Heart rate variability (HRV)
 - Sleep efficiency
 - Sleep consistency patterns
- Reduces data dimensionality while preserving important trends

5. Machine Learning & Prediction Layer

- Applies trained ML models to analyze health

patterns

- Functions:
 - Pattern recognition
 - Risk classification (low, moderate, high)
 - Early detection of lifestyle-related risks such as stress, fatigue, and cardiovascular risk
- Models are periodically retrained for better accuracy

6. Risk Assessment & Decision Layer

- Aggregates model outputs
- Generates:
 - Health risk scores
 - Personalized alerts
 - Preventive recommendations
- Supports clinical decision-making and self-monitoring

7. User Interface Layer

- Displays insights via:
 - Mobile application
 - Web dashboard
- Provides:
 - Visual health trends
 - Sleep and pulse summaries
 - Alerts and lifestyle suggestions

8. Data Storage & Security Layer

- Stores:
 - Historical health records
 - Model outputs
- Implements:
 - Secure databases
 - Access control
 - Privacy-preserving mechanisms

Predicting Lifestyle-Related Health Risks Using Pulse and Sleep Patterns



Fig 5.1: Structure of the Proposed System

VI. IMPLEMENTATION



Fig 6.1: Wearable Data Collection Screen



Fig 6.2: Raw Data Visualization Dashboard

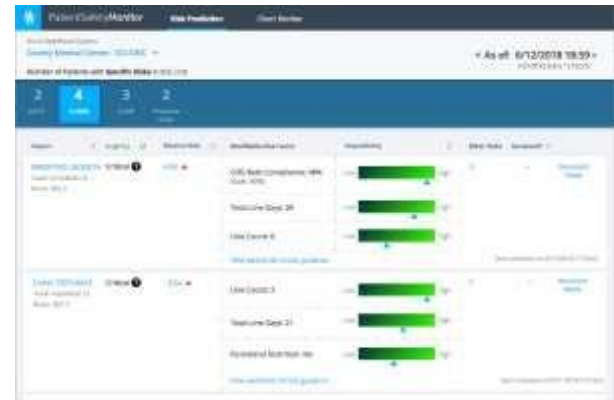


Fig 6.5: Health Risk Prediction Interface

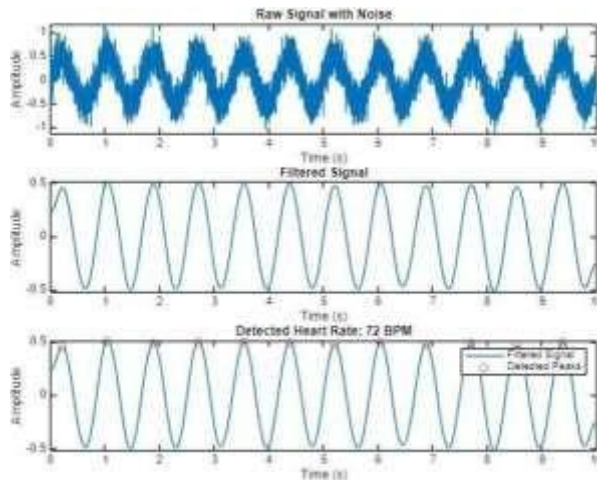


Fig 6.3: Data Preprocessing Output Screen



Fig 6.1: Alerts & Recommendation Screen



Fig 6.4: Feature Extraction Result

VII. CONCLUSION

This project presents an effective approach for predicting lifestyle-related health risks by analyzing pulse rate and sleep patterns. By leveraging data collected from wearable devices, the system is able to continuously monitor key physiological indicators that reflect an individual's daily habits and overall health status. The integration of data preprocessing, feature extraction, and machine learning-based prediction enables accurate identification of potential health risks at an early stage.

The proposed system not only classifies users into different risk levels but also provides personalized lifestyle recommendations and timely alerts, thereby supporting preventive healthcare rather than reactive treatment. The inclusion of visualization and clinician access further enhances the practical usability of the system, allowing both users and healthcare professionals to make informed decisions based on data-driven insights.

Overall, the system demonstrates how intelligent

analysis of pulse and sleep data can contribute to proactive health monitoring, improved lifestyle management, and reduced risk of chronic health conditions. With continuous model improvement and wider adoption of wearable technologies, such systems have strong potential to play a significant role in future digital healthcare solutions.

VIII. FUTURE SCOPE

The proposed system for predicting lifestyle-related health risks using pulse and sleep patterns can be further enhanced in several ways. In the future, additional physiological parameters such as blood oxygen level (SpO₂), blood pressure, physical activity intensity, stress level, and calorie expenditure can be integrated to provide a more comprehensive health assessment. Including these parameters would improve prediction accuracy and enable detection of a wider range of health conditions.

Advanced deep learning models such as LSTM, CNN-LSTM hybrids, and transformer-based time-series models can be employed to better capture long-term dependencies and complex patterns in physiological data. The system can also be extended to support real-time health monitoring with instant alerts for critical conditions, enabling faster medical intervention.

Another important future direction is the integration of the system with telemedicine platforms, allowing clinicians to remotely monitor patients and provide timely consultations. Personalized recommendation engines can be improved using reinforcement learning to adapt lifestyle suggestions based on user feedback and behavior changes over time.

Additionally, deploying the system as a scalable cloud-based or mobile application with multilingual support would increase accessibility and usability. Ensuring interoperability with diverse wearable

devices and compliance with healthcare standards will further enhance real-world adoption. Overall, these enhancements can transform the system into a robust, intelligent, and widely deployable digital health monitoring solution.

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